



Differentially Private Stochastic Gradient Descent for Outcome Weighted Learning

Aoli Yang¹ · Jun Fan² · Yunwen Lei³ · Dao-Hong Xiang¹

Received: 3 February 2026 / Revised: 14 June 2026 / Accepted: 5 September 2026
© The Author(s) 2026

Abstract

The development of individualized treatment rules in precision medicine seeks to optimize clinical outcomes for patients with varying responses to treatment. This paper focuses on outcome weighted learning, a method for estimating optimal treatment rules that take into account patient-specific characteristics within a weighted classification framework. We introduce a differentially private stochastic gradient descent algorithm within the framework of outcome weighted learning. This approach involves adding Gaussian noise to the gradient at each iteration, thereby ensuring the privacy of sensitive medical data while effectively managing large-scale datasets common in clinical practice. Unlike traditional differential privacy methods, which primarily focus on input-output type data, our approach integrates input-action-reward type data. This incorporation guarantees that both privacy and utility are preserved, with utility measured by the excess value function of the estimated individualized treatment rule. In our analysis, we establish convergence rates for the excess value function under logistic loss function, hinge loss function, and smoothed hinge loss function constructed via the Moreau envelope. These results provide rigorous utility guarantees for differentially private outcome weighted learning in precision medicine. Furthermore, empirical experiments support these theoretical findings and highlight the practical value of smoothing techniques in privacy-preserving learning for precision medicine.

Keywords Learning theory · Differential privacy · Outcome weighted learning · Stochastic gradient descent · Generalization

1 Introduction

As a groundbreaking approach in disease treatment and prevention, precision medicine tailors healthcare to the unique characteristics of individual patients. This strategy recognizes that treatment responses can vary significantly due to individual heterogeneity. A primary

Aoli Yang, Yunwen Lei, and Dao-Hong Xiang contributed equally to this work.

Editor: Lam M. Nguyen.

Extended author information available on the last page of the article

objective of precision medicine is to develop optimal individual treatment rules (ITRs) that inform treatment decisions based on specific patient attributes (Mancinelli et al., 2000). Within this innovative framework, ITRs play a crucial role in achieving optimal clinical outcomes by recognizing individual variability and customizing treatments for specific patient subgroups (Qian & Murphy, 2011), rather than adopting a one-size-fits-all methodology. In precision medicine, ensuring strong privacy protection for participants during data collection and sharing is essential, especially given the sensitive nature of clinical and medical study data. The incorporation of artificial intelligence in precision medicine has intensified discussions about patient anonymity and informed consent, highlighting the need to balance data sharing with the protection of patient information.

Differential privacy (DP) (Dwork et al., 2006; Dwork & Lei, 2009) is a privacy framework designed to protect sensitive datasets while still allowing for meaningful statistical analysis. It achieves this by adding carefully calibrated random noise to the outputs of functions derived from these datasets, effectively reducing the risk of disclosing personal information. While DP mechanisms offer strong privacy guarantees, they often involve a trade-off with data utility. A defining feature of differential privacy is its capacity to extract valuable insights about a population without compromising individual privacy. For instance, a medical database might identify a correlation between smoking and cancer, which could affect an insurance company's evaluation of a smoker's long-term medical costs. While this may result in higher premiums for the smoker, differential privacy maintains that their privacy is preserved, as the conclusions drawn from the study would remain unchanged whether or not the individual participated in the dataset. Specifically, differential privacy ensures that any sequence of outputs is "essentially" equally likely, regardless of an individual's inclusion in the dataset. This is quantified by a parameter ϵ , where smaller values indicate stronger privacy at the potential cost of accuracy. Ultimately, differential privacy provides a mathematically rigorous framework for balancing the need for data utility with the critical importance of protecting individual privacy.

Various statistical methodologies (Zhang et al., 2012) have been developed to estimate individual treatment rules. Qian and Murphy (2011) reframed the rule discovery process as an optimization challenge by introducing a value function that quantifies the average reward patients receive from their assigned treatments. However, this method prioritizes the prediction accuracy of the clinical response model rather than directly optimizing the decision rule. To overcome this limitation, Zhao et al. (2012) proposed outcome weighted learning (OWL), which frames the problem as a weighted classification problem. It is important to note that OWL differs from traditional machine learning classification. Traditional classification follows an input-output data paradigm, learning predictive correlations between features and fully observed true labels. In contrast, OWL adopts an input-action-reward data paradigm, which captures a triple-variable causal relationship and focuses on estimating causal effects. In this paradigm, the data consist of features, actual treatment actions, and rewards obtained only from the actions that were taken, while the outcomes of alternative actions remain unobserved. The treatment assignments in OWL refer to those actually assigned in the trial, rather than the ideal optimal assignments. This difference stems from the fact that labeled treatment effects are absent from the training data. The objective of OWL is to use treatment benefits to create a function that assigns individuals to their optimal treatment based on their characteristics. In addressing the challenges associated with complex electronic health record (EHR) data, Yang et al. (2025) introduced online OWL

algorithms designed to efficiently manage large-scale and streaming medical datasets. More recently, Giddens et al. (2023) proposed privacy-preserving models for ITRs, formulating OWL as a weighted empirical risk minimization problem and developing a batch DP-OWL algorithm that incorporates output perturbation. Their approach relies on full-batch gradient descent (GD) (Wang & Fan, 2024; Fan et al., 2024), which often faces high computational costs and increased memory usage, particularly when dealing with large-scale medical data such as EHR and genomic data. In contrast, stochastic gradient descent (SGD) (Bassily et al., 2014, 2019, 2020; Yang et al., 2021; Bassily et al., 2021; Chen et al., 2022) mitigates these challenges by estimating gradients from mini-batches, allowing for more frequent updates. Moreover, SGD (Feldman et al., 2020; Wang et al., 2021b, 2022; Xue et al., 2021; Su et al., 2022) is particularly well-suited for online learning environments, enabling real-time model updates and effectively handling large datasets. Consequently, in this paper, we focus on the development of differentially private outcome weighted learning algorithms by leveraging the efficiency of SGD.

This paper makes three main contributions:

- First, we present the differentially private stochastic gradient descent (DP-SGD) algorithm to estimate the optimal individualized treatment rules within the outcome weighted learning framework. Unlike traditional DP methods designed for input-output type data, our algorithm is specifically tailored for input–action–reward type data, allowing the reward to directly guide the learning of treatment rules. This algorithm adds Gaussian noise to the gradient at each iteration. By integrating differential privacy measures, our approach effectively protects medical data privacy while tackling the challenges posed by large-scale datasets, ensuring that sensitive patient information remains confidential throughout the learning process.
- Second, to address the optimization difficulties brought by non-smooth loss functions like the hinge loss function, we introduce a smooth surrogate based on the Moreau envelope technique. This smoothing technique improves training stability and allows for larger step sizes under differential privacy, while preserving the margin-based property important for ITR estimation. We further establish convergence rates with respect to the excess value function for different loss functions: while the smoothed hinge loss function matches the convergence rate of the original hinge loss function with better optimization efficiency, the logistic loss function shows slower convergence when the data distribution has more noise near the decision boundary, as characterized by the Tsybakov noise condition.
- Finally, we evaluate DP-SGD for OWL using the logistic loss function, hinge loss function, and its Moreau-smoothed variant on both simulations and real data. The results show that the empirical value generally increases as privacy constraints are relaxed, confirming that DP-SGD learns better treatment rules under looser privacy constraints. Among the loss functions considered, the Moreau-smoothed hinge loss function consistently achieves the best utility across different privacy levels. These findings support our theoretical results and highlight the practical value of smoothing for private learning in precision medicine.

This paper aims to expand the theoretical foundations of DP-SGD (Wang et al., 2022, 2024) within the OWL framework to estimate the optimal individualized treatment rules. By inte-

grating DP-SGD into OWL, we seek to enhance the privacy-preserving capabilities of the learning process while ensuring that the resulting treatment rules are both effective and tailored to individual characteristics. The remainder of this paper is structured as follows. Section 2 introduces the key concepts and framework of outcome weighted learning (OWL) and differential privacy (DP), and presents the DP-SGD algorithm tailored for OWL. Section 3 then establishes the main theoretical results, providing privacy and utility guarantees for the proposed DP-SGD algorithm. In Sect. 4, we compare our findings with existing literature, while Sect. 5 introduces our numerical study to verify the theoretical results. Finally, Sect. 6 provides detailed proofs of our main results.

2 Preliminaries

This section begins by introducing the mathematical model for individualized treatment rules. The prognostic variables for the patient are represented as $X \in \mathcal{X} \subset \mathbb{R}^d$, which are independent of the treatment assignment $A \in \mathcal{A} = \{1, -1\}$. We assume that the space of prognostic variables $\mathcal{X}$ is a bounded domain, with its diameter defined as $D_{\mathcal{X}} = \sup_{x, x' \in \mathcal{X}} \|x - x'\|_2$, where

$\|\cdot\|_2$ is the Euclidean norm. The clinical outcome for the patient, referred to as the reward, is denoted by $R \in \mathcal{R} \subseteq \mathbb{R}^+$. In general, larger values of R are viewed as more favorable.

Let P denote the joint distribution of (X, A, R) over the space $\mathcal{Z} = \mathcal{X} \times \mathcal{A} \times \mathcal{R}$, with the corresponding expectation represented as $\mathbb{E}$. An ITR is a decision rule that maps the patient's prognostic variables from $\mathcal{X}$ to the treatment space $\mathcal{A}$. The optimal ITR is defined as the one that maximizes the expected reward among all possible rules. We assume $P^{\mathcal{D}}$ represents the conditional distribution of (X, A, R) given that $A = \mathcal{D}(X)$, and that $\mathbb{E}^{\mathcal{D}}$ denotes the corresponding expectation. Assume $P(A = a) > 0$ for all $a \in \mathcal{A}$, and let $\mathbb{I}(\cdot)$ be the indicator function. Under the given treatment assignment with $\rho = P(A = 1)$, we have $P(A = a) = a\rho + (1 - a)/2$. The Radon-Nikodym derivative of $P^{\mathcal{D}}$ with respect to P is given by $\frac{dP^{\mathcal{D}}}{dP} = \frac{\mathbb{I}(a = \mathcal{D}(x))}{P(A = a)}$. Therefore, the expected reward under the ITR $\mathcal{D}$ can be calculated as follows:

$$\mathbb{E}^{\mathcal{D}}(R) = \int r dP^{\mathcal{D}} = \int r \frac{dP^{\mathcal{D}}}{dP} dP = \int r \frac{\mathbb{I}(a = \mathcal{D}(x))}{a\rho + (1 - a)/2} dP = \mathbb{E} \left[\frac{\mathbb{I}(A = \mathcal{D}(X))}{A\rho + (1 - A)/2} R \right].$$

This expectation is referred to as the value function associated with $\mathcal{D}$ and is denoted by $\mathcal{V}(\mathcal{D})$. Thus, the optimal ITR, represented as $\mathcal{D}^*$, is the rule that produces the maximum value of $\mathcal{V}(\mathcal{D})$, which can be formulated as

$$\mathcal{D}^* = \operatorname{argmax}_{\mathcal{D}} \mathbb{E} \left[\frac{\mathbb{I}(A = \mathcal{D}(X))}{A\rho + (1 - A)/2} R \right].$$

The main idea of the OWL method proposed in Zhao et al. (2012) is to reformulate the maximization problem as a minimization problem. Specifically, using the law of iterated expectations, the value function $\mathcal{V}(\mathcal{D})$ can be expressed as

$$\mathcal{V}(\mathcal{D}) = \mathbb{E}[R|A = 1] + \mathbb{E}[R|A = -1] - \mathbb{E}\left[\frac{\mathbb{I}(A \neq \mathcal{D}(X))}{A\rho + (1-A)/2}R\right],$$

which implies that

$$\mathcal{D}^* = \operatorname{argmax}_{\mathcal{D}} \mathcal{V}(\mathcal{D}) = \operatorname{argmin}_{\mathcal{D}} \mathcal{R}(\mathcal{D}), \quad (1)$$

where $\mathcal{R}(\mathcal{D}) = \mathbb{E}\left[\frac{\mathbb{I}(A \neq \mathcal{D}(X))}{A\rho + (1-A)/2}R\right]$ is referred to as the weighted misclassification error.

This approach frames the OWL method as a weighted classification problem. In this setting, as shown in Zhao et al. (2012), the optimal treatment rule $\mathcal{D}^*$ assigns treatment based on a comparison of the expected outcomes under both treatment arms. That is, $\mathcal{D}^* = \operatorname{sign}(f^*)$ with $f^*(x) = \mathbb{E}[R|X = x, A = 1] - \mathbb{E}[R|X = x, A = -1]$. Throughout the paper, we assume the reward variable R is bounded such that $0 < R \leq M$, where $M > 0$ is a constant, and define $\tilde{M} = \frac{M}{\min\{\rho, 1 - \rho\}}$.

An ITR $\mathcal{D}$ is commonly established through a real-valued function f in conjunction with the sign function, such that $\mathcal{D} = \operatorname{sign}(f)$. In this study, we assume that the target function f^* is linear. Hence, we can define the hypothesis space as

$$\mathcal{F} = \{f : f(x) = \mathbf{w}^\top x, \mathbf{w} \in \mathbb{R}^d\}.$$

The OWL optimization problem given by (1) can be expressed as $\mathcal{D}^* = \operatorname{sign}(f^*)$ with

$$f^* \in \operatorname{argmin}_{f \in \mathcal{F}} \mathbb{E}\left[\frac{\mathbb{I}(A \neq \operatorname{sign}(f(X)))}{A\rho + (1-A)/2}R\right]. \quad (2)$$

The loss function, which represents a weighted 0–1 loss function, is neither convex nor continuous. This lack of properties can render the direct solution to problem (2) potentially NP-hard and computationally impractical. To overcome this challenge, the original OWL algorithm (Zhao et al., 2012) employs a convex surrogate, the hinge loss function $\phi_h(\mathbf{a}\mathbf{w}^\top x) = \max(0, \mathbf{a}\mathbf{w}^\top x) =: (1 - \mathbf{a}\mathbf{w}^\top x)_+$, to replace the 0–1 loss function. In this paper, we explore general convex loss functions $\phi : \mathbb{R} \rightarrow \mathbb{R}_+$ that satisfy Lipschitz continuity and α -Hölder smoothness with $0 \leq \alpha \leq 1$.

Definition 1 Denote the (sub)gradient of a function g by $\partial g(\cdot)$. A function $g : \mathbf{w} \mapsto g(\mathbf{w})$ is defined to be G -Lipschitz continuous, where $G > 0$, if for all $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W} \subseteq \mathbb{R}^d$, $|g(\mathbf{w}_1) - g(\mathbf{w}_2)| \leq G\|\mathbf{w}_1 - \mathbf{w}_2\|_2$. Furthermore, a function $g : \mathbf{w} \mapsto g(\mathbf{w})$ is α -Hölder smooth, with $\alpha \in [0, 1]$ and $L > 0$, if for any $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$, $\|\partial g(\mathbf{w}_1) - \partial g(\mathbf{w}_2)\|_2 \leq L\|\mathbf{w}_1 - \mathbf{w}_2\|_2^\alpha$.

The smoothness of $g : \mathbf{w} \mapsto g(\mathbf{w})$ is characterized by the parameter $\alpha \in [0, 1]$. Specifically, $\alpha = 0$ corresponds to Lipschitz continuity, while $\alpha = 1$ indicates strong smoothness. For example, the hinge loss function $\phi_h(\mathbf{a}\mathbf{w}^\top x) = (1 - \mathbf{a}\mathbf{w}^\top x)_+$ is G -Lipschitz continuous with $G = D_x$, and it is also α -Hölder smooth with $\alpha = 0$ and $L = 2D_x$. Similarly, the logistic loss function $\phi_{\log}(\mathbf{a}\mathbf{w}^\top x) = \log(1 + e^{-\mathbf{a}\mathbf{w}^\top x})$ is G -Lipschitz continuous with $G = D_x$ and exhibits α -Hölder smoothness with $\alpha = 1$ and $L = D_x^2$.

Differential privacy (DP) is designed to address the needs of privacy-preserving data analysis. As a method that offers mathematical guarantees for privacy-preserving, DP has found widespread application across various fields (Bassily et al., 2014, 2019; Yang et al., 2021). The concept of DP is essential because it ensures that the inclusion or exclusion of a single data point does not significantly impact the results of any analysis or query, thereby protecting individual privacy. We denote two neighboring datasets S and $\tilde{S}$ by $S \sim \tilde{S}$ if $\tilde{S}$ can be obtained from S by replacing exactly one individual datum with another distinct sample from the underlying sample space.

Definition 2 (*Differential Privacy*) A randomized algorithm $\mathcal{M}$ is said to satisfy (ε, δ) -DP if, for all events $B \in \text{Range}(\mathcal{M})$ and for datasets $S \sim \tilde{S}$, the following holds

$$P(\mathcal{M}(S) \in B) \leq e^\varepsilon P(\mathcal{M}(\tilde{S}) \in B) + \delta.$$

Specifically, when we set $\delta = 0$, the algorithm is referred to as satisfying ε -DP.

Here, $\varepsilon > 0$ is the privacy budget. It bounds the maximum difference in output probabilities between two neighboring datasets. A smaller ε means this probability difference is more strictly limited, thus providing stronger privacy protection. $\delta \in (0, 1)$ is the failure probability. It describes the small probability that the differential privacy guarantee does not hold. In short, ε sets the strength of privacy protection, and δ captures rare violations of pure ε -DP.

To demonstrate that a randomized algorithm satisfies differential privacy, we introduce the below concept.

Definition 3 Let $\mathcal{W} \subseteq \mathbb{R}^d$. A function $\mathcal{G} : \mathcal{Z}^n \rightarrow \mathcal{W}$ is said to have ℓ_2 -sensitivity $\Delta_{\mathcal{G}}$ if, for all datasets $S \sim \tilde{S}$, it holds that $\|\mathcal{G}(S) - \mathcal{G}(\tilde{S})\|_2 \leq \Delta_{\mathcal{G}}$.

In this paper, we adopt the Gaussian mechanism (Dwork & Roth, 2014) to guarantee (ε, δ) -DP for the function $\mathcal{G} : \mathcal{Z}^n \rightarrow \mathcal{W}$.

Lemma 1 (See Dwork and Roth (2014)) Let $\mathcal{W} \subseteq \mathbb{R}^d$. Consider a dataset S and a function $\mathcal{G} : \mathcal{Z}^n \rightarrow \mathcal{W}$. Define $\mathbf{I}_d$ as the identity matrix in $\mathbb{R}^{d \times d}$. If the function $\mathcal{G}$ has ℓ_2 -sensitivity $\Delta_{\mathcal{G}}$, the following Gaussian mechanism ensures (ε, δ) -DP:

$$\mathcal{M}_{\mathcal{G}}(S) := \mathcal{G}(S) + \mathbf{b} \quad \text{with} \quad \mathbf{b} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d),$$

$$\text{where } \sigma = \frac{\sqrt{2\Delta_{\mathcal{G}} \log(1.25/\delta)}}{\varepsilon}.$$

In Lemma 1, the quantity σ^2 represents the variance of the Gaussian noise added to the output of the function $\mathcal{G}$. It is determined by the ℓ_2 -sensitivity of the function, the privacy budget ε , and the failure probability δ . A larger σ^2 corresponds to stronger noise, which enhances privacy protection but may degrade the utility of the output. In the context of individualized treatment rules, this utility refers to the accuracy of estimating treatment rules that maximize the value function. This highlights the fundamental trade-off between privacy and model performance in precision medicine.

Using a training dataset $S = \{(x_i, a_i, r_i)\}_{i=1}^n$ drawn independently from the distribution P on $\mathcal{Z}$, the weighted empirical risk minimization (wERM) can be defined as

$$\min_{\mathbf{w} \in \mathbb{R}^d} \left\{ \mathcal{E}_S^\phi(\mathbf{w}) := \frac{1}{n} \sum_{i=1}^n \frac{r_i}{a_i \rho + (1 - a_i)/2} \phi(a_i \mathbf{w}^\top x_i) \right\}. \quad (3)$$

To solve the wERM problem (3), we introduce the DP-SGD algorithm for OWL (Algorithm 1), which is specifically designed to address privacy concerns in sensitive medical data while accommodating large-scale datasets. The output from the algorithm applied to the dataset S is denoted as $\mathbf{w}_{\text{priv}} \in \mathbb{R}^d$. Consequently, the efficacy of this algorithm can be measured by the excess value function, defined as the difference between the optimal value function and the value function of the decision rule induced by $\mathbf{w}_{\text{priv}}$, that is, $\mathcal{V}(\mathcal{D}^*) - \mathcal{V}(\text{sign}(f_{\text{priv}}))$, where $f_{\text{priv}}(x) = \mathbf{w}_{\text{priv}}^\top x$. Note that the notation $i_t \sim \text{Unif}([n])$ means that $\{i_t\}$ is drawn uniformly from the set $[n] := \{1, 2, \dots, n\}$.

-
- 1: Inputs: Data $S = \{(x_i, a_i, r_i) \in \mathcal{Z} : i = 1, \dots, n\}$, $0 < r_i \leq M$ and $\widetilde{M} = \frac{M}{\min\{\varrho, 1-\varrho\}}$, loss function ϕ with Lipschitz parameter G , total iteration T , privacy parameters ε, δ , step size $\{\eta_t\}$ and constant β
 - 2: Set: $\mathbf{w}_1 = 0$
 - 3: **for** $t = 1$ to T **do**
 - 4: Sample $i_t \sim \text{Unif}([n])$
 - 5: $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta_t \left(\frac{r_{i_t}}{a_{i_t} \varrho + (1 - a_{i_t})/2} \partial \phi(a_{i_t} \mathbf{w}_t^\top x_{i_t}) + \mathbf{b}_t \right)$, where $\mathbf{b}_t \sim \mathcal{N}(0, \sigma^2 I_d)$
 with $\sigma^2 = \frac{14 \widetilde{M}^2 G^2 T}{\beta n^2 \varepsilon} \left(\frac{\log(1/\delta)}{(1-\beta)\varepsilon} + 1 \right)$
 - 6: **end for**
 - 7: return: $\mathbf{w}_{\text{priv}} = \frac{1}{T} \sum_{t=1}^T \mathbf{w}_t$
-

Algorithm 1 DP-SGD for OWL

3 Main Results

We use the gradient perturbation mechanism to achieve (ε, δ) -DP by adding Gaussian noise to the stochastic gradients at each iteration of the algorithm. As a result, we propose a privacy-preserving SGD algorithm tailored for OWL (Algorithm 1), which is supported by the following theorem that establishes its privacy guarantees.

Theorem 1 (Privacy Guarantee) *When the convex loss function is G -Lipschitz continuous with $G > 0$, Algorithm 1 guarantees (ε, δ) -DP for some $\beta \in (0, 1)$ if $\sigma^2 \geq 2.68 \widetilde{M}^2 G^2$ and*

$$\lambda - 1 \leq \frac{\sigma^2}{6 \widetilde{M}^2 G^2} \log \left(\frac{n}{\lambda \left(1 + \frac{\sigma^2}{4 \widetilde{M}^2 G^2} \right)} \right), \text{ where } \lambda = \frac{\log(1/\delta)}{(1-\beta)\varepsilon} + 1.$$

Remark 1 While this privacy guarantee shares a similar form with classical DP-SGD analyses (Wang et al., 2022, 2024), its underlying sensitivity analysis stems from a fundamentally different source. In traditional DP-SGD, the ℓ_2 -sensitivity is derived directly from the Lipschitz constant G of the loss function ϕ . In the OWL framework, however, the stochastic gradient (line 4 of Algorithm 1) takes the form $\frac{R}{A\rho + (1-A)/2} \partial \phi(A \mathbf{w}_t^\top X)$, where both the

reward R and the treatment assignment A are random variables. Consequently, the sensitivity of the gradient depends not only on the Lipschitz property of ϕ , but necessarily also on

the magnitude of the weighting term $\frac{R}{A\rho + (1-A)/2}$. This is precisely why the sensitiv-

ity bound in Theorem 1 involves the term $\tilde{M}G$ (with $\tilde{M} = M/\min\{\rho, 1-\rho\}$ bounding the weight). This distinction originates from the structural difference between the input-action-reward triple in OWL and the standard input-output pair in supervised learning, leading to a different privacy noise injection mechanism. The privacy proof presented here is thus re-established specifically for this weighted stochastic gradient structure.

By considering all (X, A, R) with respect to a probability measure P on $\mathcal{Z} = \mathcal{X} \times \mathcal{A} \times \mathcal{R}$, we define the weighted generalization error for a convex loss function ϕ as

$$\mathcal{E}^\phi(\mathbf{w}) = \mathbb{E} \left[\frac{R}{A\rho + (1-A)/2} \phi(A\mathbf{w}^\top X) \right] = \int_{\mathcal{Z}} \frac{r}{a\rho + (1-a)/2} \phi(a\mathbf{w}^\top x) dP,$$

and let $\mathbf{w}^* = \arg \min_{\mathbf{w} \in \mathbb{R}^d} \mathcal{E}^\phi(\mathbf{w})$. In this article, we assume that the Euclidean norm of $\mathbf{w}^*$ is bounded, that is, $\|\mathbf{w}^*\|_2 \leq B$ for some constant $B > 0$. Note that $\mathbf{w}_{\text{priv}}$ is the output of Algorithm 1, and as a first step, we assess its utility through the excess weighted generalization error $\mathcal{E}^\phi(\mathbf{w}_{\text{priv}}) - \mathcal{E}^\phi(\mathbf{w}^*)$, defined in terms of the surrogate loss function. While this quantity provides a useful optimization-level measure, our ultimate goal is to evaluate utility in terms of the excess value function associated with the learned individualized treatment rule. To bridge this gap, we introduce the necessary assumptions and theoretical tools that allow us to connect the excess weighted generalization error to the excess value function for the admissible loss functions.

Definition 4 We say that $\phi : \mathbb{R} \rightarrow \mathbb{R}_+$ is an admissible loss function if it is convex and differentiable at 0 with $\phi'(0) < 0$. Furthermore, we call ϕ a loss function with zero if its smallest zero is 1, i.e., $\phi(1) = 0$ and $\phi(t) > 0$ for all $t < 1$. We call ϕ a loss function without zero if $\phi(t) > 0$ for all $t \in \mathbb{R}$.

Assumption 1 Suppose that the marginal probability distribution $P_{\mathcal{X}}$ of P on $\mathcal{X}$ satisfies the Tsybakov noise condition with index $\tau \in [0, \infty]$, that is, there exist constants $C_1 > 0$ and $\tau \in [0, \infty]$ such that

$$P_{\mathcal{X}}(\{x \in \mathcal{X} : |f^*(x)| \leq C_1 t\}) \leq t^\tau, \text{ for all } t \geq 0. \quad (4)$$

All distributions have at least noise exponent 0 since $t^0 = 1$. The case $\tau = \infty$ means $|f^*(x)| \geq C_1$ for almost every $x \in \mathcal{X}$. Under the Tsybakov noise condition given by (4), Zhu and Xiang (2022) established a comparison theorem for admissible loss functions ϕ with zero satisfying $\phi''(0) > 0$. It is for any measurable function $f : \mathcal{X} \rightarrow \mathbb{R}$ that this theorem holds true. In particular, when considering a linear function $f(x) = \mathbf{w}^\top x$, we have

$$\mathcal{V}(\mathcal{D}^*) - \mathcal{V}(\text{sign}(f)) \leq C_1^\phi [\mathcal{E}^\phi(\mathbf{w}) - \mathcal{E}^\phi(\mathbf{w}^*)]^{\frac{\tau+1}{\tau+2}}, \quad (5)$$

where $C_1^\phi > 0$ is a constant. The above inequality can be extended to the admissible loss functions without zero under the same assumptions. We are now ready to establish an upper bound on the excess surrogate error and then extend it to a bound on the excess value func-

tion. Throughout this paper, the symbol $a_n \asymp b_n$ means that there exist positive constants C_1, C_2 such that $C_1 b_n \leq a_n \leq C_2 b_n$.

Theorem 2 Denote by $\mathbf{w}_{\text{priv}}$ the output of Algorithm 1. Assume that the convex loss function ϕ is G -Lipschitz continuous with $G > 0$ and α -Hölder smooth for some $L > 0$ and $\alpha \in [0, 1]$. If

$$\alpha < 1/2, \text{ set } T \asymp n^{\frac{2-\alpha}{1+\alpha}} \text{ and } \eta_t = \min \left\{ cn^{\frac{3(\alpha-1)}{2(1+\alpha)}}, \frac{c\varepsilon}{\sqrt{d \log(1/\delta)}} \right\} \leq \min \left\{ \frac{2}{\widetilde{ML}}, 1 \right\} \text{ for some}$$

$$\text{constant } c > 0. \text{ If } \alpha \geq 1/2, \text{ set } T \asymp n \text{ and } \eta_t = \min \left\{ \frac{c}{\sqrt{n}}, \frac{c\varepsilon}{\sqrt{d \log(1/\delta)}} \right\} \leq \min \left\{ \frac{2}{\widetilde{ML}}, 1 \right\}$$

for some constant $c > 0$. Under these conditions, we have

$$\mathbb{E} [\mathcal{E}^\phi(\mathbf{w}_{\text{priv}})] - \mathcal{E}^\phi(\mathbf{w}^*) = \mathcal{O} \left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \log(1/\delta)}}{n\varepsilon} \right).$$

Furthermore, assume that ϕ is an admissible loss function with $\phi''(0) > 0$ and the distribution P satisfies the Tsybakov noise condition (4) with $\tau \in (0, \infty]$ and $C_1 > 0$, then we have

$$\mathcal{V}(\mathcal{D}^*) - \mathbb{E} [\mathcal{V}(\text{sign}(f_{\text{priv}}))] = \mathcal{O} \left(\left(\frac{1}{\sqrt{n}} \right)^{\frac{\tau+1}{\tau+2}} + \left(\frac{\sqrt{d \log(1/\delta)}}{n\varepsilon} \right)^{\frac{\tau+1}{\tau+2}} \right).$$

Theorem 2 establishes convergence guarantees for the output $\mathbf{w}_{\text{priv}}$ under general assumptions on the convex surrogate loss function ϕ . The first part of the theorem provides a bound on the excess weighted generalization error under Lipschitz continuity and Hölder smoothness conditions. The second part further assumes that ϕ is an admissible loss function with $\phi'(0) < 0$ and $\phi''(0) > 0$, and that the data distribution satisfies the Tsybakov noise condition, in order to derive a convergence rate for the excess value function. The logistic loss $\phi_{\log}$ satisfies all the conditions of Theorem 2, including Lipschitz continuity, 1-Hölder smoothness and admissibility. The corollary below provides a theoretical guarantee on the convergence rate of the excess value function under the logistic loss function.

Corollary 1 Let $\phi = \phi_{\log}$. Assume that the distribution P satisfies the Tsybakov noise condition

$$(4) \text{ with } \tau \in (0, \infty] \text{ and } C_1 > 0. \text{ By choosing } \eta_t = \min \left\{ \frac{c}{\sqrt{n}}, \frac{c\varepsilon}{\sqrt{d \log(1/\delta)}} \right\} \leq \min \left\{ \frac{2}{\widetilde{ML}}, 1 \right\}$$

for some constant $c > 0$ and setting $T \asymp n$, we have

$$\mathcal{V}(\mathcal{D}^*) - \mathbb{E} [\mathcal{V}(\text{sign}(f_{\text{priv}}))] = \mathcal{O} \left(\left(\frac{1}{\sqrt{n}} \right)^{\frac{\tau+1}{\tau+2}} + \left(\frac{\sqrt{d \log(1/\delta)}}{n\varepsilon} \right)^{\frac{\tau+1}{\tau+2}} \right).$$

Remark 2 The convergence rate in Corollary 1 explicitly reflects the impact of the Tsybakov noise condition through the parameter $\tau \in (0, \infty]$. When τ is sufficiently large, implying

that the marginal distribution $P_{\mathcal{X}}$ has relatively few ambiguous examples in the vicinity of the decision boundary, the convergence rate approaches the optimal order of $\mathcal{O}(n^{-1/2})$. In contrast, for smaller values of τ , signifying a higher density of samples clustered near the margin, the convergence process exhibits a slower rate. This finding underscores a critical limitation of the logistic loss function $\phi_{\log}$: despite its smoothness and convexity, its gradient diminishes significantly in regions distant from the decision boundary. This gradient vanishing phenomenon can hinder optimization efficiency, especially when the data distribution places significant mass near the margin and under stringent privacy constraints (i.e., small ε values). Consequently, the logistic loss function may exhibit degraded performance in such scenarios, thereby motivating the investigation of alternative loss functions that maintain smoothness while demonstrating enhanced robustness against margin-related noise.

To further enhance the utility of differentially private learning algorithms in non-smooth settings, we consider a smoothed variant of the original loss function by using the Moreau envelope (Nesterov, 2005). This technique is especially valuable when dealing with non-smooth convex functions, as it offers a differentiable approximation that retains key convexity properties. The following definition of the ζ -Moreau envelope is originally introduced in Nesterov (2005).

Definition 5 (*Moreau envelope*) Let $g : \mathcal{W} \rightarrow \mathbb{R}$ be a convex function, and $\zeta > 0$. The ζ -Moreau envelope of g is a function $g_{\zeta} : \mathcal{W} \rightarrow \mathbb{R}$ defined as

$$g_{\zeta}(\mathbf{w}) = \min_{\mathbf{v} \in \mathcal{W}} \left(g(\mathbf{v}) + \frac{\zeta}{2} \|\mathbf{w} - \mathbf{v}\|^2 \right), \quad \mathbf{w} \in \mathcal{W}.$$

The Moreau envelope not only preserves convexity but also introduces smoothness, an essential property for applying gradient-based optimization methods under differential privacy. We summarize some of its key theoretical properties below.

Lemma 2 (See Bassily et al. (2019)) Let $g : \mathcal{W} \rightarrow \mathbb{R}$ be convex and G -Lipschitz, and $\zeta > 0$. Then the ζ -Moreau envelope $g_{\zeta}(\mathbf{w})$ satisfies the following properties:

1. g_{ζ} is convex, $2G$ -Lipschitz, and ζ -smooth.
2. For all $\mathbf{w} \in \mathcal{W}$, $g_{\zeta}(\mathbf{w}) \leq g(\mathbf{w}) \leq g_{\zeta}(\mathbf{w}) + \frac{G^2}{2\zeta}$.

In this context, we define the functions $g(\mathbf{w}) := \phi(\mathbf{a}\mathbf{w}^{\top} \mathbf{x})$ and $g_{\zeta}(\mathbf{w}) := \phi^{\zeta}(\mathbf{a}\mathbf{w}^{\top} \mathbf{x})$. The smoothing effect of the Moreau envelope allows us to more effectively apply gradient-based optimization techniques, particularly when the original loss function ϕ is non-smooth. By replacing ϕ with its Moreau envelope ϕ^{ζ} , we obtain a smooth surrogate objective. This surrogate facilitates privacy-preserving optimization via stochastic gradient descent. The subsequent result establishes a bound on the excess weighted generalization error of the DP-SGD algorithm for OWL when applied to this smoothed objective.

Theorem 3 Let $\mathbf{w}_{\text{priv}}$ be the output of Algorithm 1 run on the smoothed loss function ϕ^{ζ} , where ϕ^{ζ} is the ζ -Moreau envelope of a convex and G -Lipschitz continuous loss function ϕ with

$G > 0$. Suppose the smoothing parameter $\zeta > 0$ satisfies $\frac{MG^2}{\zeta} = \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \log(1/\delta)}}{n\varepsilon}\right)$.

Set $T \asymp n$ and $\eta_t = \min\left\{\frac{c}{\sqrt{n}}, \frac{c\varepsilon}{\sqrt{d \log(1/\delta)}}\right\} \leq \min\left\{\frac{2}{\tilde{M}\zeta}, 1\right\}$ for some constant $c > 0$.

Then we have

$$\mathbb{E}[\mathcal{E}^\phi(\mathbf{w}_{\text{priv}})] - \mathcal{E}^\phi(\mathbf{w}^*) = \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \log(1/\delta)}}{n\varepsilon}\right).$$

Theorem 3 leverages the result of Theorem 2 by applying it to the smoothed surrogate loss function ϕ^ζ , which is obtained via the ζ -Moreau envelope of the original non-smooth loss function ϕ . While ϕ^ζ is used during optimization for its smoothness, the excess weighted generalization error is still evaluated with respect to the original loss function ϕ , so that the bound reflects the model's performance on the original task.

Recall that $f_{\text{priv}}(x) = \mathbf{w}_{\text{priv}}^\top x$ and $\mathcal{D}^* = \text{sign}(f^*)$, where $f^*(x) = \mathbb{E}[R|X = x, A = 1] - \mathbb{E}[R|X = x, A = -1]$. To establish the learning rates for the excess value function $\mathcal{V}(\mathcal{D}^*) - \mathcal{V}(\text{sign}(f_{\text{priv}}))$, we further use comparison theorems within the ITR framework. These theorems show that the excess value function can be bounded by the excess weighted generalization error $\mathcal{E}^\phi(\mathbf{w}_{\text{priv}}) - \mathcal{E}^\phi(\mathbf{w}^*)$. Regarding the OWL algorithm equipped with the hinge loss function $\phi_h(u) = (1 - u)_+$, Zhao et al. (2012); Yang et al. (2025) have established that

$$\mathcal{V}(\mathcal{D}^*) - \mathcal{V}(\text{sign}(f_{\text{priv}})) \leq \mathcal{E}^{\phi_h}(\mathbf{w}_{\text{priv}}) - \mathcal{E}^{\phi_h}(\mathbf{w}^*). \quad (6)$$

The following theorem follows by combining Theorem 2, Theorem 3, and the comparison theorem for the hinge loss function.

Theorem 4 Let $\mathbf{w}_{\text{priv}}$ be the output of Algorithm 1 under different loss functions: When using the non-smooth hinge loss function ϕ_h , set the step size as

$$\eta_t = \min\left\{\frac{c}{n^{3/2}}, \frac{c\varepsilon}{\sqrt{d \log(1/\delta)}}\right\} \leq \min\left\{\frac{2}{\tilde{M}L}, 1\right\}, \text{ and the number of iterations } T \asymp n^2.$$

When using the smoothed hinge loss function ϕ_h^ζ with $\frac{MG^2}{\zeta} = \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \log(1/\delta)}}{n\varepsilon}\right)$, set

$$\eta_t = \min\left\{\frac{c}{\sqrt{n}}, \frac{c\varepsilon}{\sqrt{d \log(1/\delta)}}\right\} \leq \min\left\{\frac{2}{\tilde{M}\zeta}, 1\right\}, \text{ and } T \asymp n. \text{ Here, } c > 0 \text{ is a universal}$$

constant. In both cases, we have

$$\mathcal{V}(\mathcal{D}^*) - \mathbb{E}[\mathcal{V}(\text{sign}(f_{\text{priv}}))] = \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \log(1/\delta)}}{n\varepsilon}\right).$$

Remark 3 Theorem 4 reveals an important distinction between the non-smooth hinge loss function ϕ_h and its smoothed variant ϕ_h^ζ . Although both loss functions achieve the same convergence rate, their optimization complexities differ significantly due to the smoothness properties. In particular, the non-smooth hinge loss function requires a much smaller step size $\eta_t \asymp n^{-3/2}$ and a larger number of iterations $T \asymp n^2$ to ensure convergence under differential privacy constraints. This leads to slower convergence in practice and increased computational burden. In contrast, the smoothed hinge loss function ϕ_h^ζ , constructed via a Moreau envelope, admits larger step sizes $\eta_t \asymp n^{-1/2}$ and faster convergence with $T \asymp n$ iterations, thanks to its smoothness and controlled curvature. This highlights the theoretical advantage of using smoothed surrogates in private optimization: they enable more efficient learning without sacrificing utility guarantees. Moreover, the smooth surrogate ϕ_h^ζ maintains the margin-based behavior of the hinge loss function, which is especially beneficial for estimating individualized treatment rules under differential privacy.

4 Comparison and Discussion

There are mainly two methods to implement DP-SGD: output perturbation and gradient perturbation (Wang et al., 2022). In this paper, we employ the second method, which involves adding Gaussian noise to the gradient at each iteration. This choice prompts a review of studies on DP-SGD for traditional input-output type data learning with gradient perturbation, as it is closely related to our research.

Recently, Bassily et al. (2019) demonstrated that the mini-batch DP-SGD algorithm with gradient perturbation can ensure (ε, δ) -differential privacy (DP) and achieve optimal bound on the excess generalization error of $\mathcal{O}\left(\frac{\sqrt{d \log(1/\delta)}}{n\varepsilon} + \frac{1}{\sqrt{n}}\right)$. They addressed the

non-smoothness of the loss function using the Moreau envelope technique, which led to the optimal rate. However, their algorithms suffer from high gradient complexity, with established complexities of $\mathcal{O}\left(n^{4.5}\sqrt{\varepsilon} + \frac{n^{6.5}\varepsilon^{4.5}}{\left(d \log\left(\frac{1}{\delta}\right)\right)^2}\right)$ for non-smooth loss functions and

$\mathcal{O}\left(n^{1.5}\sqrt{\varepsilon} + \frac{(n\varepsilon)^{2.5}}{d \log\left(\frac{1}{\delta}\right)}\right)$ for strongly smooth loss functions. In contrast, Bassily et al. (2020)

presented a DP-SGD algorithm that incorporates gradient perturbation, which enhances the gradient complexity to $\mathcal{O}(n^2)$ for non-smooth loss functions. Additionally, Wang et al. (2022) established optimal utility bounds for α -Hölder smooth loss functions, achieving gradient complexities of $\mathcal{O}\left(n^{\frac{2-\alpha}{1+\alpha}} + n\right)$, which align with results from Bassily et al. (2019, 2020). However, this study only addresses cases where the parameter domain $\mathcal{W}$ is bounded. Assuming an unbounded $\mathcal{W}$ and $\mathcal{E}^\phi(\mathbf{w}^*) > 0$, Wang et al. (2024) also demonstrated that the bound on the excess generalization error of for DP-SGD achieves a rate of $\mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{1}{n\varepsilon} \sqrt{d \log(1/\delta)}\right)$ for α -Hölder smooth loss functions, with gradient complexities of $\mathcal{O}\left(n^{\frac{2-\alpha}{1+\alpha}} + n\right)$, consistent with the results in Theorem 2.

Our work extends the general DP-SGD framework of Wang et al. (2024) to the problem of learning ITRs under differential privacy, with the following key advances. First, we operate within the specific framework of the OWL algorithm, which is designed to estimate ITRs in precision medicine. This setting involves input-action-reward triplets, rather than the conventional input-output pairs assumed in standard supervised learning. As a result, the DP-SGD framework and theoretical analyses in Wang et al. (2024), which are rooted in classical learning theory, cannot be directly applied to our problem setting. Second, our analysis focuses on learning an ITR that maximizes the value function, as is standard in the OWL framework. While the surrogate weighted generalization error $\mathcal{E}^\phi(\mathbf{w})$ facilitates optimization, minimizing it does not guarantee a value-optimal decision. In particular, the idealized case where $\mathcal{E}^\phi(\mathbf{w}^*) = 0$ rarely occurs in practice and is not aligned with our primary objective. Therefore, we do not pursue the potentially faster convergence rates under this assumption, as considered in Wang et al. (2024). More importantly, our results (Theorem 3 and Theorem 4) provide explicit convergence guarantees for the excess weighted generalization error under differential privacy, even when using non-smooth loss functions such as the hinge loss function. While Wang et al. (2024) consider general α -Hölder smooth loss functions (including $\alpha = 0$), their focus is not on concrete non-smooth surrogates. In contrast, we offer a more refined analysis showing that strong bounds on excess weighted generalization error are still attainable without smoothness. These bounds are further leveraged to establish utility guarantees in terms of the excess value function. The hinge loss function, however, poses optimization challenges due to its non-smoothness, leading to gradient instability and requiring a small learning rate $\eta_t \asymp n^{-3/2}$ and a large number of iterations $T \asymp n^2$ to ensure convergence. To overcome this issue, we adopt the Moreau envelope technique to construct a smoothed surrogate of the hinge loss function (See Theorem 4). This approach enables more stable gradient behavior and more efficient optimization, while still preserving both differential privacy guarantees and utility performance.

In addition to general DP-SGD methods, several privacy-preserving approaches have been proposed for ITR estimation. Giddens et al. (2023) proposed differentially private weighted ERM and applied it to OWL. They achieved ϵ -differential privacy via output perturbation with smoothed hinge loss. Spicker et al. (2024) proposed the differentially private outcome-weighted learning framework ProWL for ITR and dynamic treatment regime (DTR) estimation. They provided ϵ -differential privacy guarantees for weighted SVM-based OWL and proved consistency of treatment assignments for clinically significant patients. Lee and Chen (2026) developed a differentially private two-stage ERM framework for ITRs. They obtained a high-probability optimization error bound, but did not derive an excess generalization error bound with explicit privacy terms. All these existing works focus on privacy guarantees, without providing complete generalization bounds or explicit utility-privacy tradeoff bounds. In contrast, our work derives a full excess generalization error bound that clearly characterizes both statistical convergence rates and privacy costs.

The conclusion of this section discusses potential future directions for our work. First, while linear models facilitate mature differential privacy guarantees and clear theoretical analysis, they may be restrictive for capturing nonlinear treatment effects. In future work, we aim to extend DP-SGD for OWL to nonlinear settings using kernel methods in reproducing kernel Hilbert spaces (RKHS) (Cucker & Zhou, 2007; Fan et al., 2019; Fan & Xiang, 2020), which further supports privacy-preserving functional learning for infinite-dimensional outputs (Jiang et al., 2023). This extension will enable us to model more complex non-

linear relationships between patient characteristics and treatment responses, significantly enhancing its practical applicability in precision medicine. Second, we plan to integrate our DP-SGD approach within federated learning frameworks (Zhang et al., 2021; Wang et al., 2021a) to facilitate collaborative model training among multiple institutions while ensuring data privacy. This integration will strengthen the generalizability of ITRs without compromising patient confidentiality.

5 Numerical Experiments

This section includes two parts: (i) simulation experiments under idealized settings to support and complement our theoretical analysis, and (ii) real-world data validation to assess the practical effectiveness of the proposed methods.

5.1 Simulation Setup

Our simulation setting is designed to replicate a standard phase II clinical trial aimed at estimating treatment effects. The prognostic variables $x_i = (x_{i,1}, x_{i,2}, x_{i,3}, x_{i,4})$ are independently drawn, where $x_{i,j} \sim U[0, 1]$ for all $i = 1, \dots, n$ and $j = 1, \dots, 4$. Each x_i is then normalized to ensure $\|x_i\|_2 = 1$. Our study focuses on binary treatments, $a \in \{-1, 1\}$, with individuals randomly assigned to either treatment with a probability of 0.5 (i.e., $\rho = 0.5$). The optimal treatment is assumed to align with $\text{sign}(f^*(x))$, where the function is defined as $f^*(x_i) = x_{i,1} + x_{i,2} - 1.8x_{i,3} - 2.2x_{i,4} + 1$. We generate the clinical outcomes r_i from a normal distribution $\mathcal{N}(\mu_i, \sigma^2)$, with μ_i defined as $0.02x_{i,4} + 3.35a_i f^*(x_i) + 0.01$ and σ set to 0.5. If any r_i is negative, we shift all benefits to $r_i + |\min\{r_i\}| + 0.001$ for $i = 1, \dots, n$ to ensure that $\frac{r_i}{a_i\rho + (1 - a_i)/2} > 0$ as required. It is anticipated that most values of r_i will fall within $(0, 15]$, with any values greater than 15 clipped to 15. This results in $\tilde{M} = 15/0.5 = 30$. Under this framework, individuals receiving the optimal treatment $\text{sign}(f^*(x))$ are expected to gain greater benefits, while those assigned to non-optimal treatments may receive lesser benefits.

We employ two methods to evaluate models trained with DP-SGD for the OWL framework. The first method assesses the average misclassification rates by comparing the estimated optimal ITR and the true optimal ITR. This is represented by $\mathbb{P}_n^*[\mathbb{I}(\text{sign}(\hat{f}(X)) \neq \text{sign}(f^*(X)))]$, where $\mathbb{P}_n^*$ denotes the empirical average, and $\hat{f}$ is the predictor function estimated by the DP-SGD for OWL. The second method estimates the expected clinical benefit $\mathcal{V}(\mathcal{D}) = \mathbb{E}\left[\frac{\mathbb{I}(A = \mathcal{D}(X))}{A\rho + (1 - A)/2}R\right]$, using the empirical treatment value defined as $\mathbb{P}_n^*[\mathbb{I}(A = \text{sign}(\hat{f}(X)))R]/\mathbb{P}_n^*[\mathbb{I}(A = \text{sign}(\hat{f}(X)))]$ as $\rho = 0.5$.

We consider three loss functions for DP-SGD in the OWL framework: the logistic loss function $\phi_{\log}$, the hinge loss function ϕ_h , and its smoothed variant ϕ_h^ζ constructed via the Moreau envelope. For the smoothed hinge loss function, we select the smoothing parameter ζ via grid search over the set $\{0.01, 0.05, 0.1, 0.2, 0.5, 1\}$, choosing the value that yields the highest empirical treatment value on training data. We conduct two sets of experiments to evaluate and compare the performance of these loss functions under differential privacy constraints.

The first experiment aims to study the impact of the privacy parameter ε on the behavior of DP-SGD for OWL using each of the three loss functions. We use sample sizes $n \in \{1000, 3000, 5000, 7000\}$, and set the number of iterations T individually for each loss function to ensure empirical convergence. The corresponding values of T are reflected in the figures. The privacy parameter ε is varied across the set $\{0.1, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, \infty\}$, where $\varepsilon = \infty$ corresponds to the non-private SGD baseline. The second experiment focuses on a fixed sample size $n = 2000$ and examines how different privacy levels $\varepsilon \in \{0.5, 2, 10, \infty\}$ affect the convergence behavior of DP-SGD with the three loss functions. For each ε , we run DP-SGD for various iteration counts $T \in \{200, 400, 600, 800, 1000, 1200, 1400, 1600\}$ to study how performance evolves over time. This setting allows us to directly compare the optimization and utility behavior of $\phi_{\log}$, ϕ_h , and ϕ_h^ζ .

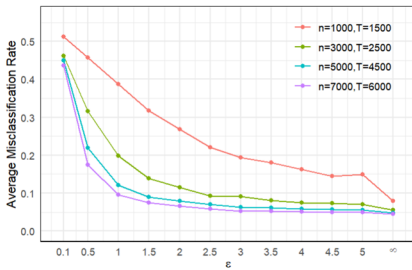
In all experiments, we fix the step size to $\eta = 0.01$, and the privacy parameter $\delta = 1/n^2$. The DP-SGD algorithm requires choosing a constant $\beta \in (0, 1)$ to ensure the (ε, δ) -DP condition. We determine the optimal β via efficient one-dimensional optimization over the feasible interval $[0.0001, 0.9999]$ to minimize the noise variance σ^2 . For each parameter combination, we repeat training 200 times and report the average misclassification rate and empirical treatment value. The results of the two experiments are presented in Figs. 1 and 2.

5.2 Simulation Results

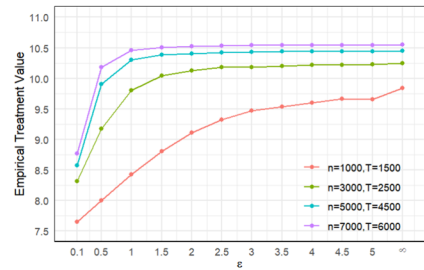
From Fig. 1, we observe that the average misclassification rates for all three loss functions consistently decrease as either the sample size n or the privacy parameter ε increases. Meanwhile, the empirical treatment values improve under the same conditions, with both metrics gradually approaching the performance of the non-private baseline (corresponding to $\varepsilon = \infty$), which aligns with our theoretical analysis (see Corollary 1 and Theorem 4). Additionally, those figures show that a larger sample size n leads to the curve reaching a plateau more quickly. This indicates that as the sample size increases, the required ε value can be lowered to achieve the same average misclassification rate and the empirical treatment value as non-private SGD for OWL. This finding has significant implications: First, a larger sample size allows the model to maintain high performance while utilizing the privacy budget more effectively, thereby enhancing privacy protection. Second, larger datasets tend to exhibit more stable statistical properties, which diminishes the influence of individual data points on model training, allowing the model to retain high efficiency even with smaller privacy perturbations. Last, the amount of noise added to DP-SGD for OWL can be reduced, as larger samples can better absorb this noise without compromising overall performance. These results advocate for the use of larger datasets in precision medicine, improving privacy protection levels without sacrificing model performance, and promoting the adoption of privacy-preserving technologies in medical applications.

The empirical results in Fig. 2 also align well with our theoretical analysis in Corollary 1 and Theorem 4. Across all privacy levels ε , the Moreau-smoothed hinge loss function ϕ_h^ζ consistently achieves the lowest misclassification rate and the highest empirical treatment value. This reflects its theoretical strengths: it preserves the convergence rate of the original hinge loss function while benefiting from smoothness, which allows for larger step sizes and faster optimization.

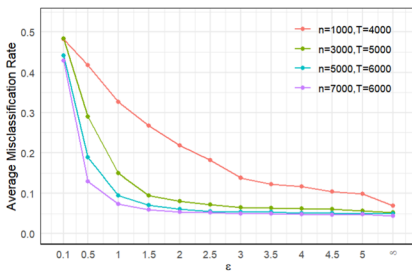
The logistic loss function $\phi_{\log}$ performs worst when the privacy budget is small (e.g., $\varepsilon = 0.5$), as the added noise severely disrupts the optimization. As ε increases, its perfor-



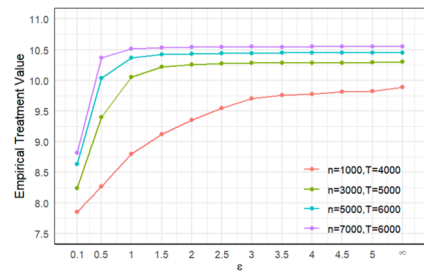
(a) Logistic Loss: Misclassification Rate



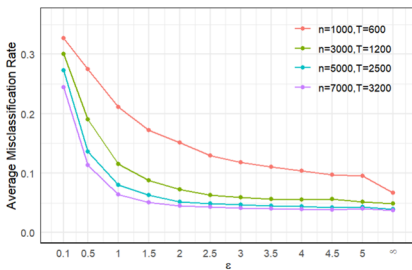
(b) Logistic Loss: Treatment Value



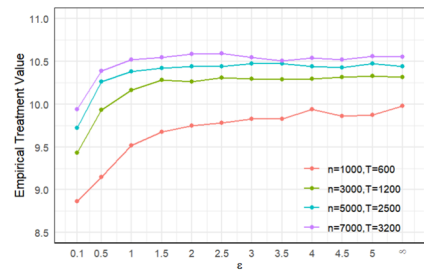
(c) Hinge Loss: Misclassification Rate



(d) Hinge Loss: Treatment Value



(e) Smooth Hinge: Misclassification Rate



(f) Smooth Hinge: Treatment Value

Fig. 1 Impact of privacy parameter ε on the performance of DP-SGD for OWL under different loss functions. Each row corresponds to one loss function: **a, b** logistic loss $\phi_{\log}$, **c, d** hinge loss ϕ_h , **e, f** smoothed hinge loss ϕ_h^ζ . Left: average misclassification rate; Right: empirical treatment value. The case $\varepsilon = \infty$ corresponds to the non-private SGD baseline

mance improves and eventually surpasses that of the non-smooth hinge loss function ϕ_h , but remains inferior to ϕ_h^ζ in all cases. This behavior is explained by Corollary 1, which shows that the convergence rate of logistic loss function depends on the Tsybakov noise exponent τ . When τ is small, a large number of samples lie close to the decision boundary, making the learning process more sensitive to any form of noise, including that introduced by privacy mechanisms. Additionally, the gradient of the logistic loss function vanishes in high-margin regions, further limiting its optimization efficiency under strong privacy constraints. In contrast, Theorem 4 shows that the convergence rates of hinge loss function and its smoothed variant are independent of τ , indicating greater robustness to label noise. This partly explains their superior performance under stringent privacy conditions.

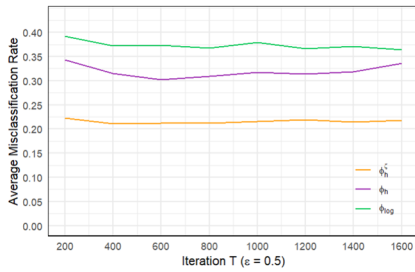
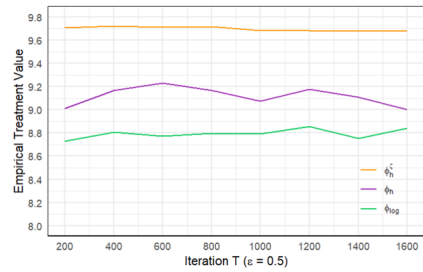
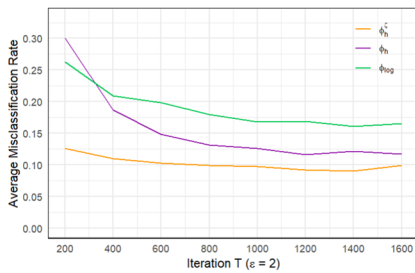
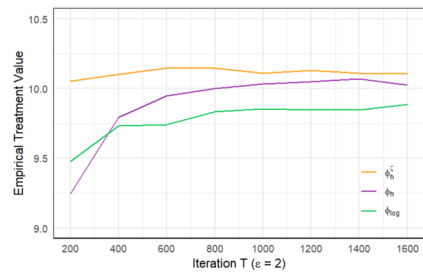
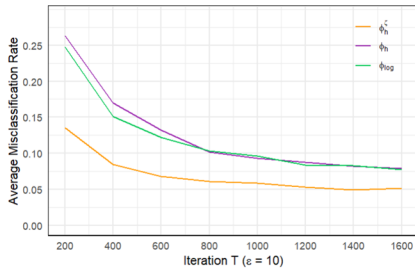
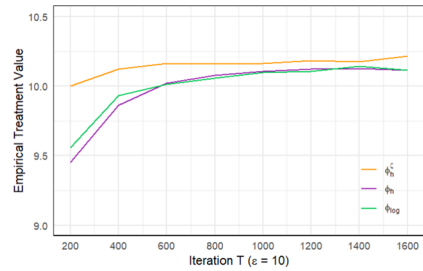
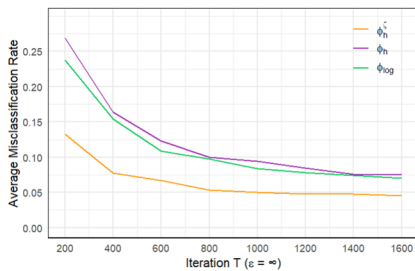
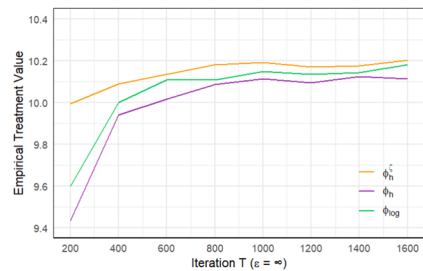
(a) $\varepsilon = 0.5$, Misclassification Rate(b) $\varepsilon = 0.5$, Treatment Value(c) $\varepsilon = 2$, Misclassification Rate(d) $\varepsilon = 2$, Treatment Value(e) $\varepsilon = 10$, Misclassification Rate(f) $\varepsilon = 10$, Treatment Value(g) $\varepsilon = \infty$, Misclassification Rate(h) $\varepsilon = \infty$, Treatment Value

Fig. 2 Comparison of loss functions ϕ_h , ϕ_h^c and $\phi_{\log}$. Average misclassification rate and empirical treatment value versus iteration count T , for fixed $n = 2000$ and varying $\varepsilon \in \{0.5, 2, 10, \infty\}$. Each row corresponds to a different value of ε , where $\varepsilon = \infty$ represents the non-private SGD baseline

The smoothed hinge loss function ϕ_h^ζ combines the margin-based property of hinge loss function with the optimization advantages of smoothness. As discussed in Remark 3, it supports larger step sizes and requires fewer iterations, improving learning efficiency under differential privacy. These theoretical and empirical results demonstrate that ϕ_h^ζ offers a favorable trade-off between utility and privacy, making it a strong choice for privacy-preserving learning of individualized treatment rules.

5.3 Validation with Real Data

In this subsection, we evaluate the practical performance of the proposed algorithms on a real-world dataset. We use the clinical trial data from the AIDS Clinical Trials Group Study 175 (ACTG 175) (Hammer et al., 1996), available from the UCI Machine Learning Repository. We select ten prognostic variables from the dataset, including seven binary variables (gender, race, homosexual activity, history of intravenous drug use, presence of symptoms, prior antiretroviral therapy, and hemophilia) and three continuous variables (age, body weight, and Karnofsky score). Before training, all continuous variables are standardized and all prognostic variables are normalized accordingly to ensure consistency across features. There are four different treatment groups: zidovudine (ZDV) monotherapy, ZDV plus didanosine (ddI), ZDV plus zalcitabine (Zal) and ddI monotherapy. The sizes of these 4 treatment groups are 532, 522, 524 and 561, respectively. Following the binary ITR setting in Qi and Liu (2018), we select four scenarios for two treatment regimes:

- Scenario 1: $A = 1$ for ZDV alone versus $A = -1$ for the other three treatments;
- Scenario 2: $A = 1$ for ZDV plus ddI versus $A = -1$ for ZDV plus Zal;
- Scenario 3: $A = 1$ for ZDV plus ddI versus $A = -1$ for ddI;
- Scenario 4: $A = 1$ for ZDV plus Zal versus $A = -1$ for ddI.

We use the change in CD4 cell count from baseline to post-treatment as the clinical outcome R , which is a direct and reliable biomarker for evaluating HIV treatment effect (Fu et al., 2019; Qi & Liu, 2018). Another common choice in existing literature is survival time, which is subject to censoring (Choi et al., 2023; Jiang et al., 2017).

To evaluate models trained using DP-SGD within the OWL framework, we adopt the following empirical treatment value:

$$\mathcal{V}(\hat{f}) = \frac{\mathbb{P}_n^* \left[\frac{\mathbb{I}(A=\hat{f}(X))}{A\rho+(1-A)/2} R \right]}{\mathbb{P}_n^* \left[\frac{\mathbb{I}(A=\hat{f}(X))}{A\rho+(1-A)/2} \right]},$$

where $\hat{f}$ denotes the treatment decision function learned by the proposed model. We compare three loss functions: the logistic loss $\phi_{\log}$, the hinge loss ϕ_h and a smoothed variant of the hinge loss ϕ_h^ζ with $\zeta = 1$, obtained via the Moreau envelope. We fix the number of training iterations at $T = 1000$ and the step size at $\eta = 0.1$ for all methods. We adopt 5-fold cross-validation to mitigate the randomness of data partitioning. Each experiment is repeated 200 times to account for randomness, and we report the mean and standard deviation (SD) across runs. The results are summarized in Table 1. Table 1 reveals two important patterns across all four scenarios. First, as the privacy parameter ε increases, the average treatment

Table 1 Comparison of empirical treatment values across selected ε for loss functions ϕ_h^ζ , ϕ_h , and $\phi_{\log}$ in four different data cases

	ε	ϕ_h^ζ	ϕ_h	$\phi_{\log}$
Scenario 1	0.1	7.120 (5.677)	6.756 (5.354)	6.936 (5.279)
	0.5	11.998 (5.331)	11.731 (5.131)	11.368 (5.556)
	1.5	19.732 (3.424)	19.436 (3.740)	19.704 (3.365)
	2.5	22.609 (2.735)	22.246 (2.618)	22.499 (2.656)
	3.5	24.209 (2.104)	23.788 (2.245)	23.971 (2.255)
	4.5	24.849 (1.765)	24.660 (2.055)	24.875 (1.652)
	5.5	25.238 (1.576)	24.856 (2.097)	24.975 (1.802)
	6.5	25.524 (1.442)	25.313 (1.618)	25.163 (1.532)
	7.5	25.484 (1.359)	25.284 (1.786)	25.347 (1.526)
	8.5	25.623 (1.345)	25.531 (1.595)	25.490 (1.440)
	9.5	25.769 (1.373)	25.643 (1.367)	25.632 (1.384)
	∞	25.985 (1.353)	25.853 (1.214)	25.784 (1.310)
Scenario 2	0.1	31.784 (4.909)	31.092 (4.002)	31.673 (4.596)
	0.5	32.824 (4.551)	32.455 (4.911)	32.095 (4.555)
	1.5	34.613 (4.277)	34.502 (4.387)	34.747 (3.980)
	2.5	35.986 (4.021)	35.658 (3.839)	35.840 (4.015)
	3.5	36.721 (3.609)	36.605 (3.967)	37.246 (3.730)
	4.5	37.609 (3.779)	37.296 (3.648)	37.219 (3.325)
	5.5	38.111 (3.696)	37.643 (3.517)	37.801 (3.903)
	6.5	38.967 (3.538)	37.666 (3.485)	38.536 (3.363)
	7.5	38.897 (3.662)	38.794 (3.313)	38.769 (3.440)
	8.5	38.979 (3.204)	38.709 (3.526)	38.702 (3.203)
	9.5	38.944 (3.320)	38.775 (3.190)	38.731 (3.644)
	∞	40.316 (2.913)	39.926 (2.758)	39.931 (2.931)
Scenario 3	0.1	37.523 (4.400)	37.206 (4.030)	36.962 (4.222)
	0.5	37.812 (4.357)	37.611 (4.028)	37.661 (3.955)
	1.5	38.528 (4.226)	38.474 (4.150)	38.389 (3.442)
	2.5	38.697 (3.681)	38.681 (3.926)	38.727 (4.183)
	3.5	38.858 (3.950)	38.797 (3.588)	38.867 (4.116)
	4.5	39.153 (3.991)	39.567 (4.122)	39.674 (3.951)
	5.5	39.951 (3.540)	39.795 (3.840)	39.128 (3.640)
	6.5	40.390 (3.783)	39.841 (3.336)	39.714 (3.538)
	7.5	40.403 (3.778)	40.129 (4.140)	39.977 (3.816)
	8.5	40.368 (3.841)	40.152 (3.898)	40.670 (3.546)
	9.5	41.174 (3.331)	40.051 (3.678)	40.377 (3.517)
	∞	42.521 (3.745)	42.050 (3.207)	42.270 (3.339)

Table 1 (continued)

	ε	ϕ_h^ζ	ϕ_h	$\phi_{\log}$
Scenario 4	0.1	21.915 (3.256)	21.790 (3.719)	21.558 (3.469)
	0.5	22.141 (3.215)	22.089 (3.286)	22.138 (3.210)
	1.5	22.280 (3.202)	22.187 (3.507)	22.183 (3.700)
	2.5	22.469 (3.710)	22.452 (3.359)	22.531 (3.093)
	3.5	22.463 (3.414)	22.615 (3.372)	22.560 (3.305)
	4.5	22.849 (3.222)	22.643 (3.389)	22.866 (3.230)
	5.5	22.871 (3.294)	22.727 (3.333)	22.883 (3.423)
	6.5	22.912 (3.149)	22.737 (3.363)	22.811 (3.214)
	7.5	23.049 (3.157)	22.840 (3.405)	22.922 (3.177)
	8.5	23.932 (3.421)	22.921 (3.290)	22.805 (3.406)
	9.5	23.022 (3.061)	22.928 (3.084)	22.848 (3.456)
	∞	23.322 (3.102)	23.245 (2.844)	23.097 (3.178)

All values are reported as mean (SD) across repeated runs. $\varepsilon = \infty$ corresponds to the non-private SGD baseline

values for all three loss functions ϕ_h^ζ , ϕ_h and $\phi_{\log}$ exhibit a general upward trend. This confirms that the DP-SGD algorithm is able to learn better treatment rules under looser privacy constraints, consistent with theoretical and simulation results. Second, across all scenarios and privacy levels, across all privacy levels, the smoothed hinge loss function ϕ_h^ζ consistently yields the highest average treatment value. This highlights its empirical advantage and effectiveness in optimizing treatment decisions under differential privacy constraints.

6 Proofs of Main Results

6.1 Proof of Privacy Guarantee

In this section, we present the detailed proofs of our main results. To establish Theorem 1, we begin with the definition of Rényi differential privacy (RDP), which offers a more refined framework for privacy analysis than traditional DP. RDP allows for a tighter analysis of composition and amplification mechanisms. This concept was first introduced in Mironov (2017).

Definition 6 Let $\lambda > 1$ and $\kappa > 0$. Consider a randomized algorithm $\mathcal{M}$. If the following inequality holds for all $S \sim \tilde{S}$:

$$D_\lambda(\mathcal{M}(S), \mathcal{M}(\tilde{S})) := \frac{1}{\lambda - 1} \log \int \left(\frac{P_{\mathcal{M}(S)}(\theta)}{P_{\mathcal{M}(\tilde{S})}(\theta)} \right)^\lambda dP_{\mathcal{M}(\tilde{S})}(\theta) \leq \kappa,$$

where $P_{\mathcal{M}(S)}(\theta)$ is the density of $\mathcal{M}(S)$, then $\mathcal{M}$ is said to satisfy (λ, κ) -RDP.

We now introduce a lemma that outlines essential RDP properties to support our analysis, particularly focusing on the privacy amplification effect of RDP via uniform subsampling (Liang et al., 2020).

Lemma 3 Let $\mathcal{W} \subseteq \mathbb{R}^d$ and $\mathcal{G} : \mathcal{Z}^n \rightarrow \mathcal{W}$ be a function. We denote the ℓ_2 -sensitivity of $\mathcal{G}$ by $\Delta_{\mathcal{G}}$. Given a dataset $S \subset \mathcal{Z}^n$ of size n , the Gaussian mechanism for $\mathcal{G}$ is defined as $\mathcal{M}_{\mathcal{G}}(S) = \mathcal{G}(S) + \mathbf{b}$, where $\mathbf{b}$ is drawn from a Gaussian distribution $\mathcal{N}(0, \sigma^2 \mathbf{I}_d)$. Let S_{sub} be a randomly chosen subset of S , selected uniformly from all subsets of size k . The subsampling rate is denoted as $\gamma = k/n$. If $\sigma^2 \geq 0.67 \Delta_{\mathcal{G}}^2$ and $\lambda - 1 \leq \frac{2\sigma^2}{3\Delta_{\mathcal{G}}^2} \log\left(\frac{1}{\lambda\gamma(1 + \sigma^2/\Delta_{\mathcal{G}}^2)}\right)$, then applying $\mathcal{M}_{\mathcal{G}}$ to S_{sub} satisfies $(\lambda, 3.5\gamma^2 \lambda \Delta_{\mathcal{G}}^2 / \sigma^2)$ -RDP.

Based on the adaptive composition theorem (Mironov, 2017) related to RDP, the privacy of a sequence of adaptive mechanisms can be evaluated by analyzing the privacy guarantees of each individual mechanism.

Lemma 4 (Adaptive Composition of RDP) A mechanism $\mathcal{M}$ that comprises adaptive mechanisms $(\mathcal{M}_1, \dots, \mathcal{M}_t)$, each satisfying (λ, κ_k) -RDP for $k \in [t]$, is itself $(\lambda, \sum_{k=1}^t \kappa_k)$ -RDP.

Next, we describe the relationship between RDP and DP (Mironov, 2017).

Lemma 5 (From RDP to (ϵ, δ) -DP) Given that $\mathcal{M}$ is (λ, κ) -RDP, it follows that $\mathcal{M}$ is $(\kappa + \log(1/v)/(\lambda - 1), v)$ -DP for all $v \in (0, 1)$.

Another key characteristic of differential privacy is its post-processing property (Mironov, 2017).

Lemma 6 (Post-processing) Let $\mathcal{W}_1, \mathcal{W}_2 \subseteq \mathbb{R}^d$. If $\mathcal{M} : \mathcal{Z}^n \rightarrow \mathcal{W}_1$ satisfies (λ, κ) -RDP, then for any function $f : \mathcal{W}_1 \rightarrow \mathcal{W}_2$, the composition $f \circ \mathcal{M} : \mathcal{Z}^n \rightarrow \mathcal{W}_2$ also satisfies (λ, κ) -RDP.

Building on the lemmas presented above, we outline the proof of Theorem 1, which relates to the privacy guarantees of Algorithm 1.

Proof of Theorem 1 We begin by confirming that the output of each iteration $\mathbf{w}_t$ satisfies RDP, ensuring that $\mathbf{w}_{\text{priv}}$ meets the requirements for differential privacy. For all $t \in [T]$, we

analyze the mechanism $\mathcal{M}_t = \mathcal{G}_t + \mathbf{b}_t$, where $\mathcal{G}_t = \frac{r_{i_t}}{a_{i_t}\rho + (1 - a_{i_t})/2} \partial\phi(a_{i_t} \mathbf{w}_t^\top x_{i_t})$. From the Lipschitz continuity of ϕ , we can deduce that for any $z_{i_t}, z'_{i_t} \in \mathcal{Z}$,

$$\left\| \frac{r_{i_t}}{a_{i_t}\rho + (1 - a_{i_t})/2} \partial\phi(a_{i_t} \mathbf{w}_t^\top x_{i_t}) - \frac{r'_{i_t}}{a'_{i_t}\rho + (1 - a'_{i_t})/2} \partial\phi(a'_{i_t} \mathbf{w}_t^\top x'_{i_t}) \right\|_2 \leq \tilde{M} \left\| \partial\phi(a_{i_t} \mathbf{w}_t^\top x_{i_t}) \right\|_2 + \tilde{M} \left\| \partial\phi(a'_{i_t} \mathbf{w}_t^\top x'_{i_t}) \right\|_2 \leq 2\tilde{M}G. \quad (7)$$

From Definition 3, it follows that the ℓ_2 -sensitivity of $\mathcal{G}_t$ is $2\tilde{M}G$. We note that

$$\sigma^2 = \frac{14\tilde{M}^2 G^2 T}{\beta n^2 \varepsilon} \left(\frac{\log(1/\delta)}{(1-\beta)\varepsilon} + 1 \right).$$

Applying Lemma 3 with $\gamma = 1/n$, we have that $\mathcal{M}_t$ is $\left(\lambda, \frac{\lambda\beta\varepsilon}{T \left(\frac{\log(1/\sigma)}{(1-\beta)\varepsilon} + 1 \right)} \right)$ -RDP under the conditions

$$\frac{\sigma^2}{\tilde{M}^2 G^2} \geq 2.68 \quad (8)$$

and

$$\lambda - 1 \leq \frac{\sigma^2}{6\tilde{M}^2 G^2} \log \left(\frac{n}{\lambda \left(1 + \frac{\sigma^2}{4\tilde{M}^2 G^2} \right)} \right). \quad (9)$$

Thus, Lemma 6 implies that $\mathbf{w}_{t+1}$ satisfies $\left(\lambda, \frac{\lambda\beta\varepsilon}{T \left(\frac{\log(1/\sigma)}{(1-\beta)\varepsilon} + 1 \right)} \right)$ -RDP. Furthermore, by applying Lemma 4 to the output after T iterations, we demonstrate that Algorithm 1 satisfies $\left(\lambda, \frac{\lambda\beta\varepsilon}{T \left(\frac{\log(1/\delta)}{(1-\beta)\varepsilon} + 1 \right)} \right)$ -RDP. Selecting $\lambda = \frac{\log(1/\delta)}{(1-\beta)\varepsilon} + 1$ allows us to conclude that Algorithm 1 satisfies the $\left(\frac{\log(1/\delta)}{(1-\beta)\varepsilon} + 1, \beta\varepsilon \right)$ -RDP condition. Therefore, Lemma 5, along with (8) and (9), leads to the desired conclusion. $\square$

6.2 Error Analysis

For simplicity, we will use $\mathcal{E}$ and $\mathcal{E}_S$ in place of $\mathcal{E}^\phi$ and $\mathcal{E}_S^\phi$ in the subsequent discussion. In this subsection, we examine the convergence rate of the excess value function for DP-SGD in the context of OWL. To achieve this, we need to establish the utility guarantee of

$\mathbf{w}_{\text{priv}} = \frac{1}{T} \sum_{t=1}^T \mathbf{w}_t$, which relies on the error decomposition given by:

$$\begin{aligned} \mathbb{E} [\mathcal{E}(\mathbf{w}_{\text{priv}}) - \mathcal{E}(\mathbf{w}^*)] &\leq \frac{1}{T} \sum_{t=1}^T \mathbb{E} [\mathcal{E}(\mathbf{w}_t) - \mathcal{E}(\mathbf{w}^*)] \\ &= \frac{1}{T} \sum_{t=1}^T \mathbb{E} [\mathcal{E}(\mathbf{w}_t) - \mathcal{E}_S(\mathbf{w}_t)] + \frac{1}{T} \sum_{t=1}^T \mathbb{E} [\mathcal{E}_S(\mathbf{w}_t) - \mathcal{E}_S(\mathbf{w}^*)], \end{aligned}$$

where the inequality follows from Jensen's inequality, and the equality follows from $\mathbb{E}[\mathcal{E}_S(\mathbf{w}^*)] = \mathcal{E}(\mathbf{w}^*)$. The first term, $\frac{1}{T} \sum_{t=1}^T \mathbb{E} [\mathcal{E}(\mathbf{w}_t) - \mathcal{E}_S(\mathbf{w}_t)]$, is the generalization error. To bound this term, we will employ algorithmic stability (Bassily et al., 2020; Bousquet &

Elisseeff, 2002; Hardt et al., 2016; Kuzborskij & Lampert, 2018; Lei & Ying, 2020). The

second term, $\frac{1}{T} \sum_{t=1}^T \mathbb{E} [\mathcal{E}_S(\mathbf{w}_t) - \mathcal{E}_S(\mathbf{w}^*)]$, represents the optimization error, which can be

addressed using techniques from optimization theory. The proof of Theorem 2 is inspired by the analytical framework proposed in Wang et al. (2024), which separates the generalization error and the optimization error.

6.2.1 Estimate Generalization Error

First, we will establish a bound on the generalization error using the definition of on-average model stability (Fan & Lei, 2024; Lei & Ying, 2020). Essentially, this definition measures the average sensitivity of models by analyzing perturbations in individual coordinates.

Definition 7 Let $z_i, \tilde{z}_i$ be drawn independently from P . Define $S = \{z_i\}_{i=1}^n$, $\tilde{S} = \{\tilde{z}_i\}_{i=1}^n$, and $S^{(i)} = \{z_1, \dots, z_{i-1}, \tilde{z}_i, z_{i+1}, \dots, z_n\}$ for any $i \in [n]$. We define $\mathcal{M}$ to be ℓ_2 on-average model v -stable if the following condition holds

$$\mathbb{E}_{S, \tilde{S}, \mathcal{M}} \left[\frac{1}{n} \sum_{i=1}^n \left\| \mathcal{M}(S) - \mathcal{M}(S^{(i)}) \right\|_2^2 \right] \leq v.$$

Let $c_{\alpha,1}$ be a constant that depends only on α , defined as follows

$$c_{\alpha,1} = \begin{cases} (1 + \frac{1}{\alpha})^{\frac{\alpha}{1+\alpha}} L^{\frac{1}{1+\alpha}}, & \text{if } \alpha > 0, \\ \phi(0) + L, & \text{if } \alpha = 0. \end{cases} \quad (10)$$

In this subsection, we will explore how the generalization error is related to ℓ_2 on-average model stability.

Lemma 7 Let $\kappa > 0$. Assume that the convex loss function ϕ is α -Hölder smooth for some $L > 0$ and $\alpha \in [0, 1]$. If $\mathcal{M}$ is ℓ_2 on-average model v -stable, then we have

$$\mathbb{E}_{S, \mathcal{M}} [\mathcal{E}(\mathcal{M}(S)) - \mathcal{E}_S(\mathcal{M}(S))] \leq \frac{c_{\alpha,1}^2 \tilde{M}^{\frac{1-\alpha}{1+\alpha}}}{2\kappa} \mathbb{E}_{S, \mathcal{M}} \left[\mathcal{E}^{\frac{2\alpha}{1+\alpha}}(\mathcal{M}(S)) \right] + \frac{\kappa \tilde{M} v}{2}. \quad (11)$$

Proof Due to the symmetry in the data, we obtain

$$\begin{aligned} & \mathbb{E}_{S, \mathcal{M}} [\mathcal{E}(\mathcal{M}(S)) - \mathcal{E}_S(\mathcal{M}(S))] \\ &= \mathbb{E}_{S, \tilde{S}, \mathcal{M}} \left[\frac{1}{n} \sum_{i=1}^n (\mathcal{E}(\mathcal{M}(S^{(i)})) - \mathcal{E}_S(\mathcal{M}(S))) \right] \\ &= \mathbb{E}_{S, \tilde{S}, \mathcal{M}} \left\{ \frac{1}{n} \sum_{i=1}^n \left[\frac{r_i}{a_i \rho + (1 - a_i)/2} \left(\phi(a_i \mathcal{M}(S^{(i)})^\top x_i) - \phi(a_i \mathcal{M}(S)^\top x_i) \right) \right] \right\}. \end{aligned}$$

Here the second equality follows from the fact that $\mathcal{M}(S^{(i)})$ is independent of $\mathbf{z}_i$. From this, along with the convexity of ϕ , we can deduce

$$\begin{aligned} & \mathbb{E}_{S, \mathcal{M}} [\mathcal{E}(\mathcal{M}(S)) - \mathcal{E}_S(\mathcal{M}(S))] \\ & \leq \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{S, \mathcal{S}, \mathcal{M}} \left[\frac{r_i}{a_i \rho + (1 - a_i)/2} \langle \mathcal{M}(S^{(i)}) - \mathcal{M}(S), \partial \phi(a_i \mathcal{M}(S^{(i)})^\top x_i) \rangle \right]. \end{aligned} \quad (12)$$

According to Lemma A.1 in Lei and Ying (2020), for a map $\mathbf{w} \mapsto g(\mathbf{w}; z)$ that is nonnegative and α -Hölder smooth for all $z \in \mathcal{Z}$, there holds

$$\|\partial g(\mathbf{w}; z)\|_2 \leq c_{\alpha, 1} g^{\frac{\alpha}{1+\alpha}}(\mathbf{w}; z), \quad (13)$$

for all $\mathbf{w} \in \mathbb{R}^d$ and $z \in \mathcal{Z}$. This together with Cauchy-Schwartz inequality implies

$$\begin{aligned} & \langle \mathcal{M}(S^{(i)}) - \mathcal{M}(S), \partial \phi(a_i \mathcal{M}(S^{(i)})^\top x_i) \rangle \\ & \leq \frac{\kappa}{2} \|\mathcal{M}(S^{(i)}) - \mathcal{M}(S)\|_2^2 + \frac{1}{2\kappa} \|\partial \phi(a_i \mathcal{M}(S^{(i)})^\top x_i)\|_2^2 \\ & \leq \frac{\kappa}{2} \|\mathcal{M}(S^{(i)}) - \mathcal{M}(S)\|_2^2 + \frac{c_{\alpha, 1}^2}{2\kappa} \phi^{\frac{2\alpha}{1+\alpha}}(a_i \mathcal{M}(S^{(i)})^\top x_i). \end{aligned} \quad (14)$$

Combining (12) and (14), we derive

$$\begin{aligned} & \mathbb{E}_{S, \mathcal{M}} [\mathcal{E}(\mathcal{M}(S)) - \mathcal{E}_S(\mathcal{M}(S))] \\ & \leq \frac{\tilde{M}\kappa}{2n} \sum_{i=1}^n \mathbb{E}_{S, \mathcal{S}, \mathcal{M}} \left[\|\mathcal{M}(S^{(i)}) - \mathcal{M}(S)\|_2^2 \right] \\ & \quad + \frac{c_{\alpha, 1}^2 \tilde{M}^{\frac{1-\alpha}{1+\alpha}}}{2\kappa} \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{S, \mathcal{S}, \mathcal{M}} \left[\left(\frac{r_i}{a_i \rho + (1 - a_i)/2} \phi(a_i \mathcal{M}(S^{(i)})^\top x_i) \right)^{\frac{2\alpha}{1+\alpha}} \right], \end{aligned} \quad (15)$$

where the inequality follows from the assumption that $0 < R \leq M$ and $\tilde{M} = \frac{M}{\min\{\rho, 1 - \rho\}}$.

Given the concavity of $x \mapsto x^{\frac{2\alpha}{1+\alpha}}$ and the independence of $\mathbf{z}_i$ from $A(S^{(i)})$, we can deduce

$$\begin{aligned} & \mathbb{E}_{S, \mathcal{S}, \mathcal{M}} \left[\left(\frac{r_i}{a_i \rho + (1 - a_i)/2} \phi(a_i \mathcal{M}(S^{(i)})^\top x_i) \right)^{\frac{2\alpha}{1+\alpha}} \right] \\ & \leq \mathbb{E}_{S, \mathcal{S}, \mathcal{M}} \left[\left(\mathbb{E}_{\mathbf{z}_i} \left[\frac{r_i}{a_i \rho + (1 - a_i)/2} \phi(a_i \mathcal{M}(S^{(i)})^\top x_i) \right] \right)^{\frac{2\alpha}{1+\alpha}} \right] \\ & = \mathbb{E}_{S, \mathcal{S}, \mathcal{M}} \left[\mathcal{E}^{\frac{2\alpha}{1+\alpha}}(\mathcal{M}(S^{(i)})) \right] \\ & = \mathbb{E}_{S, \mathcal{M}} \left[\mathcal{E}^{\frac{2\alpha}{1+\alpha}}(\mathcal{M}(S)) \right]. \end{aligned} \quad (16)$$

Therefore, substituting (16) into (15) leads us to our conclusion. $\square$

To establish the stability bounds for Algorithm 1 when dealing with α -Hölder smooth loss functions, we rely on two lemmas as follows.

Lemma 8 Assume that for some $L > 0$ and $\alpha \in [0, 1]$, the convex loss function ϕ is α -Hölder smooth. It follows that

$$\begin{aligned} & \left\langle \mathbf{w} - \tilde{\mathbf{w}}, \frac{r}{a\rho + (1-a)/2} (\partial\phi(a\mathbf{w}^\top x) - \partial\phi(a\tilde{\mathbf{w}}^\top x)) \right\rangle \\ & \geq \frac{2(\tilde{M}L)^{-\frac{1}{\alpha}}}{1+\alpha} \left\| \frac{r}{a\rho + (1-a)/2} (\partial\phi(a\mathbf{w}^\top x) - \partial\phi(a\tilde{\mathbf{w}}^\top x)) \right\|_2^{\frac{1+\alpha}{\alpha}}, \end{aligned}$$

holds for all $\mathbf{w}, \tilde{\mathbf{w}} \in \mathbb{R}^d$.

Proof According to Lemma D.2 in Lei and Ying (2020), for a map $\mathbf{w} \mapsto g(\mathbf{w}; z)$ that is convex and α -Hölder smooth for all $z \in \mathcal{Z}$, the inequality

$$\langle \mathbf{w} - \tilde{\mathbf{w}}, \partial g(\mathbf{w}; z) - \partial g(\tilde{\mathbf{w}}; z) \rangle \geq \frac{2L_1^{-\frac{1}{\alpha}}}{1+\alpha} \|\partial g(\mathbf{w}; z) - \partial g(\tilde{\mathbf{w}}; z)\|_2^{\frac{1+\alpha}{\alpha}}$$

holds for any $\mathbf{w}, \tilde{\mathbf{w}} \in \mathbb{R}^d$ and all $z \in \mathcal{Z}$. We define $g(\mathbf{w}; z) = \frac{r}{a\rho + (1-a)/2} \phi(a\mathbf{w}^\top x)$. Given the α -Hölder smoothness of ϕ , we have

$$\begin{aligned} & \left\| \partial \left(\frac{r}{a\rho + (1-a)/2} \phi(a\mathbf{w}^\top x) \right) - \partial \left(\frac{r}{a\rho + (1-a)/2} \phi(a\tilde{\mathbf{w}}^\top x) \right) \right\|_2 \\ & = \left\| \frac{r}{a\rho + (1-a)/2} (\partial\phi(a\mathbf{w}^\top x) - \partial\phi(a\tilde{\mathbf{w}}^\top x)) \right\|_2 \\ & \leq \tilde{M}L \|\mathbf{w} - \tilde{\mathbf{w}}\|_2^\alpha, \end{aligned} \quad (17)$$

which shows that g is α -Hölder smooth with parameters $\tilde{M}L$ and $\alpha \in [0, 1]$. Thus, the proof is complete. $\square$

Let $c_{\alpha, \tilde{M}, 1}$ be a constant that depends on α and $\tilde{M}$, defined as follows

$$c_{\alpha, \tilde{M}, 1} = \sqrt{\frac{1-\alpha}{1+\alpha}} (2^{-\alpha} \tilde{M}L)^{\frac{1}{1-\alpha}}. \quad (18)$$

The following lemma provides a bound on the expansive behavior of the mapping $\mathbf{w} \mapsto \mathbf{w} - \eta \frac{r}{a\rho + (1-a)/2} \partial\phi(a\mathbf{w}^\top x)$ for convex loss function ϕ with Hölder continuous (sub)gradients.

Lemma 9 Let $\eta > 0$. If the convex loss function ϕ is α -Hölder smooth for some $L > 0$ and $\alpha \in [0, 1]$, then for all $z \in \mathcal{Z}$ and $\mathbf{w} \in \mathbb{R}^d$, we have

$$\left\| \mathbf{w} - \eta \frac{r}{a\rho + (1-a)/2} \partial\phi(a\mathbf{w}^\top x) - \tilde{\mathbf{w}} + \eta \frac{r}{a\rho + (1-a)/2} \partial\phi(a\tilde{\mathbf{w}}^\top x) \right\|_2^2 \leq \|\mathbf{w} - \tilde{\mathbf{w}}\|_2^2 + c_{\alpha, \tilde{M}, 1}^2 \eta^{\frac{2}{1-\alpha}}.$$

Proof It can be shown that the following equality holds

$$\begin{aligned} & \left\| \mathbf{w} - \eta \frac{r}{a\rho + (1-a)/2} \partial\phi(a\mathbf{w}^\top x) - \tilde{\mathbf{w}} + \eta \frac{r}{a\rho + (1-a)/2} \partial\phi(a\tilde{\mathbf{w}}^\top x) \right\|_2^2 \\ &= \|\mathbf{w} - \tilde{\mathbf{w}}\|_2^2 + \eta^2 \left\| \frac{r}{a\rho + (1-a)/2} [\partial\phi(a\mathbf{w}^\top x) - \partial\phi(a\tilde{\mathbf{w}}^\top x)] \right\|_2^2 \\ &\quad - 2\eta \left\langle \mathbf{w} - \tilde{\mathbf{w}}, \frac{r}{a\rho + (1-a)/2} [\partial\phi(a\mathbf{w}^\top x) - \partial\phi(a\tilde{\mathbf{w}}^\top x)] \right\rangle. \end{aligned} \quad (19)$$

First, we consider the case where $\alpha = 0$. Here, the α -Hölder smooth of ϕ along with Lemma 8 for $\alpha = 0$ leads to

$$\left\| \mathbf{w} - \eta \frac{r}{a\rho + (1-a)/2} \partial\phi(a\mathbf{w}^\top x) - \tilde{\mathbf{w}} + \eta \frac{r}{a\rho + (1-a)/2} \partial\phi(a\tilde{\mathbf{w}}^\top x) \right\|_2^2 \leq \|\mathbf{w} - \tilde{\mathbf{w}}\|_2^2 + 4\eta^2(\tilde{M}L)^2.$$

Next, we examine the case where $\alpha > 0$. Using Lemma 8, we can conclude that

$$\begin{aligned} & \left\| \frac{r}{a\rho + (1-a)/2} [\partial\phi(a\mathbf{w}^\top x) - \partial\phi(a\tilde{\mathbf{w}}^\top x)] \right\|_2^2 \\ & \leq \left(\frac{(\tilde{M}L)^{\frac{1}{\alpha}}(1+\alpha)}{2\alpha} \left\langle \mathbf{w} - \tilde{\mathbf{w}}, \frac{r}{a\rho + (1-a)/2} [\partial\phi(a\mathbf{w}^\top x) - \partial\phi(a\tilde{\mathbf{w}}^\top x)] \right\rangle \right)^{\frac{2\alpha}{1+\alpha}} \\ &= \left(\frac{1+\alpha}{\eta\alpha} \left\langle \mathbf{w} - \tilde{\mathbf{w}}, \frac{r}{a\rho + (1-a)/2} [\partial\phi(a\mathbf{w}^\top x) - \partial\phi(a\tilde{\mathbf{w}}^\top x)] \right\rangle \right)^{\frac{2\alpha}{1+\alpha}} \left(\eta^{\frac{2\alpha}{1+\alpha}} (\tilde{M}L)^{\frac{2}{1+\alpha}} 2^{-\frac{2\alpha}{1+\alpha}} \right) \\ & \leq \frac{2\alpha}{1+\alpha} \left(\left(\frac{1+\alpha}{\eta\alpha} \right)^{\frac{2\alpha}{1+\alpha}} \left\langle \mathbf{w} - \tilde{\mathbf{w}}, \frac{r}{a\rho + (1-a)/2} [\partial\phi(a\mathbf{w}^\top x) - \partial\phi(a\tilde{\mathbf{w}}^\top x)] \right\rangle \right)^{\frac{2\alpha}{1+\alpha}} \\ & \quad + \frac{1-\alpha}{1+\alpha} \left(\eta^{\frac{2\alpha}{1+\alpha}} (\tilde{M}L)^{\frac{2}{1+\alpha}} 2^{-\frac{2\alpha}{1+\alpha}} \right)^{\frac{1+\alpha}{1-\alpha}} \\ &= 2\eta^{-1} \left\langle \mathbf{w} - \tilde{\mathbf{w}}, \frac{r}{a\rho + (1-a)/2} [\partial\phi(a\mathbf{w}^\top x) - \partial\phi(a\tilde{\mathbf{w}}^\top x)] \right\rangle + \frac{1-\alpha}{1+\alpha} \eta^{\frac{2\alpha}{1-\alpha}} (2^{-\alpha}\tilde{M}L)^{\frac{2}{1-\alpha}}. \end{aligned} \quad (20)$$

Here, we have used Young's inequality (Cucker & Zhou, 2007)

$$ab \leq p^{-1}|a|^p + q^{-1}|b|^q, \text{ for } a, b \in \mathbb{R}, p, q > 0 \text{ with } p^{-1} + q^{-1} = 1. \quad (21)$$

Substituting inequality (20) back into (19) gives

$$\begin{aligned} & \left\| \mathbf{w} - \eta \frac{r}{a\rho + (1-a)/2} \partial\phi(a\mathbf{w}^\top x) - \tilde{\mathbf{w}} + \eta \frac{r}{a\rho + (1-a)/2} \partial\phi(a\tilde{\mathbf{w}}^\top x) \right\|_2^2 \\ & \leq \|\mathbf{w} - \tilde{\mathbf{w}}\|_2^2 + \frac{1-\alpha}{1+\alpha} \eta^{2+\frac{2\alpha}{1-\alpha}} (2^{-\alpha}\tilde{M}L)^{\frac{2}{1-\alpha}}. \end{aligned}$$

We derive the required bound by combining the two cases. The proof is complete. $\square$

To analyze the stability of DP-SGD for OWL, we assess how adding a consistent Gaussian noise sequence to the stochastic gradient affects neighboring datasets. Consequently, using the two aforementioned lemmas, we establish the stability bounds for Algorithm 1 when applied to α -Hölder smooth loss functions.

Lemma 10 Define $\mathcal{M}(S)$ as the $(t+1)$ -th iteration of Algorithm 1 on the dataset S . If the convex loss function ϕ is α -Hölder smooth for some $L > 0$ and $\alpha \in [0, 1]$, then $\mathcal{M}$ is ℓ_2 on-average model v -stable with

$$v \leq \frac{4ec_{\alpha,1}^2(1+t/n)\tilde{M}^{\frac{2}{1+\alpha}}}{n} \sum_{j=1}^t \eta_j^2 \mathbb{E} \left[\mathcal{E}_S^{\frac{2\alpha}{1+\alpha}}(\mathbf{w}_j) \right] + c_{\alpha,\tilde{M},1}^2 e \sum_{j=1}^t \eta_j^{\frac{2}{1-\alpha}}. \quad (22)$$

Proof When $i_t \neq i$, Lemma 9 leads to

$$\begin{aligned} & \|\mathbf{w}_{t+1} - \mathbf{w}_{t+1}^{(i)}\|_2^2 \\ &= \left\| \mathbf{w}_t - \eta_t \frac{r_{i_t}}{a_{i_t}\rho + (1-a_{i_t})/2} \partial\phi(a_{i_t}\mathbf{w}_t^\top x_{i_t}) - \mathbf{w}_t^{(i)} + \eta_t \frac{r_{i_t}}{a_{i_t}\rho + (1-a_{i_t})/2} \partial\phi(a_{i_t}\mathbf{w}_t^{(i)\top} x_{i_t}) \right\|_2^2 \\ &\leq \|\mathbf{w}_t - \mathbf{w}_t^{(i)}\|_2^2 + c_{\alpha,\tilde{M},1}^2 \eta_t^{\frac{2}{1-\alpha}}. \end{aligned}$$

When $i_t = i$, we have

$$\begin{aligned} & \|\mathbf{w}_{t+1} - \mathbf{w}_{t+1}^{(i)}\|_2^2 \\ &= \left\| \mathbf{w}_t - \eta_t \frac{r_i}{a_i\rho + (1-a_i)/2} \partial\phi(a_i\mathbf{w}_t^\top x_i) - \mathbf{w}_t^{(i)} + \eta_t \frac{\tilde{r}_i}{\tilde{a}_i\rho + (1-\tilde{a}_i)/2} \partial\phi(\tilde{a}_i\mathbf{w}_t^{(i)\top} \tilde{x}_i) \right\|_2^2 \\ &\leq (p+1)\|\mathbf{w}_t - \mathbf{w}_t^{(i)}\|_2^2 + 2(p^{-1}+1)\eta_t^2 \left(\left\| \frac{r_i}{a_i\rho + (1-a_i)/2} \partial\phi(a_i\mathbf{w}_t^\top x_i) \right\|_2^2 \right. \\ &\quad \left. + \left\| \frac{\tilde{r}_i}{\tilde{a}_i\rho + (1-\tilde{a}_i)/2} \partial\phi(\tilde{a}_i\mathbf{w}_t^{(i)\top} \tilde{x}_i) \right\|_2^2 \right), \end{aligned}$$

where we applied the inequality $(a+b)^2 \leq (p+1)a^2 + (1/p+1)b^2$. By combining the two inequalities above, we take into account the distribution of i_t and use (13), leading to

$$\begin{aligned} \mathbb{E}_{\mathcal{M}} \left[\|\mathbf{w}_{t+1} - \mathbf{w}_{t+1}^{(i)}\|_2^2 \right] &\leq (p/n+1) \left(\mathbb{E}_{\mathcal{M}} \left[\|\mathbf{w}_t - \mathbf{w}_t^{(i)}\|_2^2 \right] + c_{\alpha,\tilde{M},1}^2 \eta_t^{\frac{2}{1-\alpha}} \right) \\ &\quad + \frac{2(p^{-1}+1)c_{\alpha,1}^2 \eta_t^2 \tilde{M}^{\frac{2}{1+\alpha}}}{n} \mathbb{E}_{\mathcal{M}} \left[\left(\frac{r_i}{a_i\rho + (1-a_i)/2} \phi(a_i\mathbf{w}_t^\top x_i) \right)^{\frac{2\alpha}{1+\alpha}} \right. \\ &\quad \left. + \left(\frac{\tilde{r}_i}{\tilde{a}_i\rho + (1-\tilde{a}_i)/2} \phi(\tilde{a}_i\mathbf{w}_t^{(i)\top} \tilde{x}_i) \right)^{\frac{2\alpha}{1+\alpha}} \right]. \end{aligned}$$

The identical distribution of z_i and $\tilde{z}_i$ allows us to conclude that

$$\mathbb{E}_{S, \tilde{S}, \mathcal{M}} \left[\left(\frac{\tilde{r}_i}{\tilde{a}_i \rho + (1 - \tilde{a}_i)/2} \phi(\tilde{a}_i \mathbf{w}_t^\top \tilde{x}_i) \right)^{\frac{2\alpha}{1+\alpha}} \right] = \mathbb{E}_{S, \mathcal{M}} \left[\left(\frac{r_i}{a_i \rho + (1 - a_i)/2} \phi(a_i \mathbf{w}_t^\top x_i) \right)^{\frac{2\alpha}{1+\alpha}} \right].$$

Thus, we have

$$\begin{aligned} & \mathbb{E}_{S, \tilde{S}, \mathcal{M}} \left[\|\mathbf{w}_{t+1} - \mathbf{w}_{t+1}^{(i)}\|_2^2 \right] \\ & \leq (p/n + 1) \left(\mathbb{E}_{S, \tilde{S}, \mathcal{M}} \left[\|\mathbf{w}_t - \mathbf{w}_t^{(i)}\|_2^2 \right] + c_{\alpha, \tilde{M}, 1}^2 \eta_t^{\frac{2}{1-\alpha}} \right) \\ & \quad + \frac{4(p^{-1} + 1) c_{\alpha, 1}^2 \eta_t^2 \tilde{M}^{\frac{2}{1+\alpha}}}{n} \mathbb{E}_{S, \mathcal{M}} \left[\left(\frac{r_i}{a_i \rho + (1 - a_i)/2} \phi(a_i \mathbf{w}_t^\top x_i) \right)^{\frac{2\alpha}{1+\alpha}} \right]. \end{aligned}$$

Multiplying both sides by $(p/n + 1)^{-(t+1)}$, we obtain

$$\begin{aligned} & (p/n + 1)^{-(t+1)} \mathbb{E}_{S, \tilde{S}, \mathcal{M}} \left[\|\mathbf{w}_{t+1} - \mathbf{w}_{t+1}^{(i)}\|_2^2 \right] \\ & \leq (p/n + 1)^{-t} \left(\mathbb{E}_{S, \tilde{S}, \mathcal{M}} \left[\|\mathbf{w}_t - \mathbf{w}_t^{(i)}\|_2^2 \right] + c_{\alpha, \tilde{M}, 1}^2 \eta_t^{\frac{2}{1-\alpha}} \right) \\ & \quad + \frac{4(p^{-1} + 1) c_{\alpha, 1}^2 (p/n + 1)^{-(t+1)} \eta_t^2 \tilde{M}^{\frac{2}{1+\alpha}}}{n} \mathbb{E}_{S, \mathcal{M}} \left[\left(\frac{r_i}{a_i \rho + (1 - a_i)/2} \phi(a_i \mathbf{w}_t^\top x_i) \right)^{\frac{2\alpha}{1+\alpha}} \right]. \end{aligned}$$

Taking the sum over j , starting with $\mathbf{w}_1 = \mathbf{w}_1^{(i)} = 0$, we obtain

$$\begin{aligned} & (p/n + 1)^{-(t+1)} \mathbb{E}_{S, \tilde{S}, \mathcal{M}} \left[\|\mathbf{w}_{t+1} - \mathbf{w}_{t+1}^{(i)}\|_2^2 \right] \\ & \leq c_{\alpha, \tilde{M}, 1}^2 \sum_{j=1}^t (1 + p/n)^{-j} \eta_j^{\frac{2}{1-\alpha}} \\ & \quad + \frac{4(p^{-1} + 1) c_{\alpha, 1}^2 \tilde{M}^{\frac{2}{1+\alpha}}}{n} \sum_{j=1}^t (p/n + 1)^{-(j+1)} \eta_j^2 \mathbb{E}_{S, \mathcal{M}} \left[\left(\frac{r_i}{a_i \rho + (1 - a_i)/2} \phi(a_i \mathbf{w}_j^\top x_i) \right)^{\frac{2\alpha}{1+\alpha}} \right]. \end{aligned}$$

By averaging over i , we find

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{S, \tilde{S}, \mathcal{M}} \left[\|\mathbf{w}_{t+1} - \mathbf{w}_{t+1}^{(i)}\|_2^2 \right] \leq c_{\alpha, \tilde{M}, 1}^2 \sum_{j=1}^t (p/n + 1)^{t+1-j} \eta_j^{\frac{2}{1-\alpha}} \\ & \quad + \frac{4(1 + p^{-1}) c_{\alpha, 1}^2 \tilde{M}^{\frac{2}{1+\alpha}}}{n^2} \sum_{j=1}^t \sum_{i=1}^n (p/n + 1)^{t-j} \eta_j^2 \mathbb{E}_{S, \mathcal{M}} \left[\left(\frac{r_i}{a_i \rho + (1 - a_i)/2} \phi(a_i \mathbf{w}_j^\top x_i) \right)^{\frac{2\alpha}{1+\alpha}} \right]. \end{aligned}$$

Since the map $x \mapsto x^{\frac{2\alpha}{1+\alpha}}$ is concave, we obtain by Jensen's inequality that

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{S, \mathcal{S}, \mathcal{M}} \left[\|\mathbf{w}_{t+1} - \mathbf{w}_{t+1}^{(i)}\|_2^2 \right] &\leq c_{\alpha, \widetilde{M}, 1}^2 \sum_{j=1}^t (p/n + 1)^{t+1-j} \eta_j^{\frac{2}{1-\alpha}} \\ &+ 4(p^{-1} + 1) c_{\alpha, 1}^2 \widetilde{M}^{\frac{2}{1+\alpha}} \sum_{j=1}^t \frac{(p/n + 1)^{t-j} \eta_j^2}{n} \mathbb{E}_{S, \mathcal{M}} \left[\left(\frac{1}{n} \sum_{i=1}^n \frac{r_i}{a_i \rho + (1 - a_i)/2} \phi(a_i \mathbf{w}_j^\top x_i) \right)^{\frac{2\alpha}{1+\alpha}} \right]. \end{aligned}$$

Consequently, the inequality holds based on the definition of $\mathcal{E}_S$, completing our proof. $\square$

Using Lemma 7 and Lemma 10, we can directly derive Proposition 1, which plays a crucial role in establishing the generalization bound for DP-SGD in the context of OWL.

Proposition 1 *Let $\kappa > 0$. If there exist $L > 0$ and $\alpha \in [0, 1]$ such that the convex loss function ϕ is α -Hölder smooth, then we have*

$$\begin{aligned} \mathbb{E} [\mathcal{E}(\mathbf{w}_{t+1}) - \mathcal{E}_S(\mathbf{w}_{t+1})] &\leq \frac{c_{\alpha, 1}^2 \widetilde{M}^{\frac{1-\alpha}{1+\alpha}}}{2\kappa} \mathbb{E} \left[\mathcal{E}^{\frac{2\alpha}{1+\alpha}}(\mathbf{w}_{t+1}) \right] \\ &+ \frac{\kappa}{2} \left(e \widetilde{M} c_{\alpha, \widetilde{M}, 1}^2 \sum_{j=1}^t \eta_j^{\frac{2}{1-\alpha}} + \frac{4e(t/n + 1) c_{\alpha, 1}^2 \widetilde{M}^{\frac{3+\alpha}{1+\alpha}}}{n} \sum_{j=1}^t \eta_j^2 \mathbb{E} \left[\mathcal{E}_S^{\frac{2\alpha}{1+\alpha}}(\mathbf{w}_j) \right] \right), \end{aligned}$$

where $c_{\alpha, 1}$ and $c_{\alpha, \widetilde{M}, 1}$ are defined in (10) and (18), respectively.

6.2.2 Estimate Optimization Error

In this subsection, we outline the optimization error associated with convex loss functions that exhibit α -Hölder smoothness. Recall that $\mathbf{w}^* = \arg \min_{\mathbf{w} \in \mathbb{R}^d} \mathcal{E}(\mathbf{w})$. We introduce the constant

$$c_{\alpha, \widetilde{M}, 2} = \begin{cases} c_{\alpha, 1}^2 \widetilde{M}^2, & \text{if } \alpha = 0, \\ \frac{1-\alpha}{1+\alpha} (2\alpha/(1+\alpha))^{\frac{2\alpha}{1-\alpha}} c_{\alpha, 1}^{\frac{2+2\alpha}{1-\alpha}} \widetilde{M}^{\frac{2}{1-\alpha}}, & \text{if } \alpha > 0. \end{cases}$$

Proposition 2 *Let $\kappa > 0$. Suppose the step size η_j is non-increasing and that $\{\mathbf{w}_j\}$ is generated by Algorithm 1. If there exist $L > 0$ and $\alpha \in [0, 1]$ such that the convex loss function ϕ is α -Hölder smooth, then the following holds*

$$\begin{aligned} &\sum_{j=1}^t \eta_j \mathbb{E} [\mathcal{E}_S(\mathbf{w}_j) - \mathcal{E}_S(\mathbf{w}^*)] \\ &\leq \frac{1}{2} \|\mathbf{w}^*\|_2^2 + \frac{3}{2} \sum_{j=1}^t \eta_j^2 \sigma^2 d + \frac{3}{4} c_{\alpha, 1}^2 \widetilde{M}^{\frac{2}{1+\alpha}} \left(\sum_{j=1}^t \eta_j^2 \right)^{\frac{1-\alpha}{1+\alpha}} \left[2\eta_1 \|\mathbf{w}^*\|_2^2 \right. \\ &\quad \left. + \sum_{j=1}^t (6\eta_j^3 \sigma^2 d + 4\eta_j^2 \mathcal{E}(\mathbf{w}^*) + 3c_{\alpha, \widetilde{M}, 2} \eta_j^{\frac{3-\alpha}{1-\alpha}}) \right]^{\frac{2\alpha}{1+\alpha}}. \end{aligned}$$

Proof From the definition of $\mathbf{w}_{t+1}$, we can observe that for every $\alpha \in [0, 1]$,

$$\begin{aligned}
 \|\mathbf{w}_{t+1} - \mathbf{w}^*\|_2^2 &= \left\| \mathbf{w}_t - \eta_t \left(\frac{r_{i_t}}{a_{i_t}\rho + (1-a_{i_t})/2} \partial\phi(a_{i_t}\mathbf{w}_t^\top x_{i_t}) + \mathbf{b}_t \right) - \mathbf{w}^* \right\|_2^2 \\
 &= \|\mathbf{w}_t - \mathbf{w}^*\|_2^2 + \eta_t^2 \left\| \frac{r_{i_t}}{a_{i_t}\rho + (1-a_{i_t})/2} \partial\phi(a_{i_t}\mathbf{w}_t^\top x_{i_t}) + \mathbf{b}_t \right\|_2^2 \\
 &\quad + 2\eta_t \left\langle \mathbf{w}^* - \mathbf{w}_t, \frac{r_{i_t}}{a_{i_t}\rho + (1-a_{i_t})/2} \partial\phi(a_{i_t}\mathbf{w}_t^\top x_{i_t}) + \mathbf{b}_t \right\rangle \\
 &\leq \|\mathbf{w}_t - \mathbf{w}^*\|_2^2 + \frac{3}{2}\eta_t^2 \left\| \frac{r_{i_t}}{a_{i_t}\rho + (1-a_{i_t})/2} \partial\phi(a_{i_t}\mathbf{w}_t^\top x_{i_t}) \right\|_2^2 + 3\eta_t^2 \|\mathbf{b}_t\|_2^2 \\
 &\quad + 2\eta_t \left\langle \mathbf{w}^* - \mathbf{w}_t, \frac{r_{i_t}}{a_{i_t}\rho + (1-a_{i_t})/2} \partial\phi(a_{i_t}\mathbf{w}_t^\top x_{i_t}) + \mathbf{b}_t \right\rangle \\
 &\leq \|\mathbf{w}_t - \mathbf{w}^*\|_2^2 + \frac{3}{2}c_{\alpha,1}^2\eta_t^2\tilde{M}^{\frac{2}{1+\alpha}} \left(\frac{r_{i_t}}{a_{i_t}\rho + (1-a_{i_t})/2} \phi(a_{i_t}\mathbf{w}_t^\top x_{i_t}) \right)^{\frac{2\alpha}{1+\alpha}} + 3\eta_t^2 \|\mathbf{b}_t\|_2^2 \\
 &\quad + 2\eta_t \frac{r_{i_t}}{a_{i_t}\rho + (1-a_{i_t})/2} [\phi(a_{i_t}\mathbf{w}^{*\top} x_{i_t}) - \phi(a_{i_t}\mathbf{w}_t^\top x_{i_t})] + 2\eta_t \langle \mathbf{w}^* - \mathbf{w}_t, \mathbf{b}_t \rangle.
 \end{aligned} \tag{23}$$

Here we apply $(a+b)^2 \leq \frac{3a^2}{2} + 3b^2$ in the first inequality, and the second inequality follows from (13) and the convexity of ϕ . Thus, we obtain

$$\begin{aligned}
 2\eta_t \frac{r_{i_t}}{a_{i_t}\rho + (1-a_{i_t})/2} [\phi(a_{i_t}\mathbf{w}_t^\top x_{i_t}) - \phi(a_{i_t}\mathbf{w}^{*\top} x_{i_t})] &\leq \|\mathbf{w}_t - \mathbf{w}^*\|_2^2 - \|\mathbf{w}_{t+1} - \mathbf{w}^*\|_2^2 \\
 + \frac{3}{2}c_{\alpha,1}^2\eta_t^2\tilde{M}^{\frac{2}{1+\alpha}} \left(\frac{r_{i_t}}{a_{i_t}\rho + (1-a_{i_t})/2} \phi(a_{i_t}\mathbf{w}_t^\top x_{i_t}) \right)^{\frac{2\alpha}{1+\alpha}} &+ 3\eta_t^2 \|\mathbf{b}_t\|_2^2 + 2\eta_t \langle \mathbf{w}^* - \mathbf{w}_t, \mathbf{b}_t \rangle.
 \end{aligned}$$

Summing over j and noting that $\mathbf{w}_1 = \mathbf{0}$, we derive

$$\begin{aligned}
 &2 \sum_{j=1}^t \eta_j \frac{r_{i_j}}{a_{i_j}\rho + (1-a_{i_j})/2} [\phi(a_{i_j}\mathbf{w}_j^\top x_{i_j}) - \phi(a_{i_j}\mathbf{w}^{*\top} x_{i_j})] \\
 &\leq \|\mathbf{w}^*\|_2^2 + \frac{3}{2}c_{\alpha,1}^2\tilde{M}^{\frac{2}{1+\alpha}} \sum_{j=1}^t \eta_j^2 \left(\frac{r_{i_j}}{a_{i_j}\rho + (1-a_{i_j})/2} \phi(a_{i_j}\mathbf{w}_j^\top x_{i_j}) \right)^{\frac{2\alpha}{1+\alpha}} \\
 &\quad + \sum_{j=1}^t \left(3\eta_j^2 \|\mathbf{b}_j\|_2^2 + 2\eta_j \langle \mathbf{w}^* - \mathbf{w}_j, \mathbf{b}_j \rangle \right).
 \end{aligned}$$

Since $\mathbf{w}_j$ is independent of i_j , we have

$$\begin{aligned}
 \sum_{j=1}^t \eta_j \mathbb{E} [\mathcal{E}_S(\mathbf{w}_j) - \mathcal{E}_S(\mathbf{w}^*)] &= \sum_{j=1}^t \eta_j \mathbb{E} \left[\frac{r_{i_j}}{a_{i_j}\rho + (1-a_{i_j})/2} (\phi(a_{i_j}\mathbf{w}_j^\top x_{i_j}) - \phi(a_{i_j}\mathbf{w}^{*\top} x_{i_j})) \right] \\
 &\leq \frac{1}{2} \|\mathbf{w}^*\|_2^2 + \frac{3}{4}c_{\alpha,1}^2\tilde{M}^{\frac{2}{1+\alpha}} \sum_{j=1}^t \eta_j^2 \mathbb{E} \left[\left(\frac{r_{i_j}}{a_{i_j}\rho + (1-a_{i_j})/2} \phi(a_{i_j}\mathbf{w}_j^\top x_{i_j}) \right)^{\frac{2\alpha}{1+\alpha}} \right] + \frac{3}{2} \sum_{j=1}^t \eta_j^2 \sigma^2 d,
 \end{aligned} \tag{24}$$

where we use the fact that $\mathbb{E}[\|\mathbf{b}_j\|_2^2] = \sigma^2 d$ and $\mathbb{E}[\langle \mathbf{w}^* - \mathbf{w}_j, \mathbf{b}_j \rangle] = 0$. This holds true because $\mathbf{w}^* - \mathbf{w}_j$ is independent of $\mathbf{b}_j$, which is a Gaussian vector with zero mean and variance σ^2 .

Now we are in the position to estimate

$$\sum_{j=1}^t \eta_j^2 \mathbb{E} \left[\left(\frac{r_{ij}}{a_{ij}\rho + (1-a_{ij})/2} \phi(a_{ij} \mathbf{w}_j^\top x_{ij}) \right)^{\frac{2\alpha}{1+\alpha}} \right].$$

By applying Young's inequality (21), we find that for any $t \in [T]$,

$$\begin{aligned} & \eta_t c_{\alpha,1}^2 \tilde{M}^{\frac{2}{1+\alpha}} \left(\frac{r_{it}}{a_{it}\rho + (1-a_{it})/2} \phi(a_{it} \mathbf{w}_t^\top x_{it}) \right)^{\frac{2\alpha}{1+\alpha}} \\ &= \left(\frac{1+\alpha}{2\alpha} \frac{r_{it}}{a_{it}\rho + (1-a_{it})/2} \phi(a_{it} \mathbf{w}_t^\top x_{it}) \right)^{\frac{2\alpha}{1+\alpha}} \left(\frac{2\alpha}{1+\alpha} \right)^{\frac{2\alpha}{1+\alpha}} c_{\alpha,1}^2 \tilde{M}^{\frac{2}{1+\alpha}} \eta_t \\ &\leq \frac{2\alpha}{1+\alpha} \left(\frac{1+\alpha}{2\alpha} \frac{r_{it}}{a_{it}\rho + (1-a_{it})/2} \phi(a_{it} \mathbf{w}_t^\top x_{it}) \right) + \frac{1-\alpha}{1+\alpha} \left(\left(\frac{2\alpha}{1+\alpha} \right)^{\frac{2\alpha}{1+\alpha}} c_{\alpha,1}^2 \tilde{M}^{\frac{2}{1+\alpha}} \eta_t \right)^{\frac{1+\alpha}{1-\alpha}} \\ &= \frac{r_{it}}{a_{it}\rho + (1-a_{it})/2} \phi(a_{it} \mathbf{w}_t^\top x_{it}) + c_{\alpha, \tilde{M}, 2} \eta_t^{\frac{1+\alpha}{1-\alpha}}. \end{aligned}$$

Incorporating this inequality into (23) gives us

$$\begin{aligned} \|\mathbf{w}_{t+1} - \mathbf{w}^*\|_2^2 &\leq \|\mathbf{w}_t - \mathbf{w}^*\|_2^2 + 3\eta_t^2 \|\mathbf{b}_t\|_2^2 + 2\eta_t \frac{r_{it}}{a_{it}\rho + (1-a_{it})/2} \phi(a_{it} \mathbf{w}^{*\top} x_{it}) \\ &\quad - \frac{1}{2}\eta_t \frac{r_{it}}{a_{it}\rho + (1-a_{it})/2} \phi(a_{it} \mathbf{w}_t^\top x_{it}) + \frac{3}{2} c_{\alpha, \tilde{M}, 2} \eta_t^{\frac{2}{1-\alpha}} + 2\eta_t \langle \mathbf{w}^* - \mathbf{w}_t, \mathbf{b}_t \rangle. \end{aligned}$$

By simple calculation, it follows that

$$\begin{aligned} & \eta_t^2 \frac{r_{it}}{a_{it}\rho + (1-a_{it})/2} \phi(a_{it} \mathbf{w}_t^\top x_{it}) \\ &\leq 2\eta_t \|\mathbf{w}_t - \mathbf{w}^*\|_2^2 - 2\eta_t \|\mathbf{w}_{t+1} - \mathbf{w}^*\|_2^2 + 6\eta_t^3 \|\mathbf{b}_t\|_2^2 + 4\eta_t^2 \frac{r_{it}}{a_{it}\rho + (1-a_{it})/2} \phi(a_{it} \mathbf{w}^{*\top} x_{it}) \\ &\quad + 3c_{\alpha, \tilde{M}, 2} \eta_t^{\frac{3-\alpha}{1-\alpha}} + 4\eta_t^2 \langle \mathbf{w}^* - \mathbf{w}_t, \mathbf{b}_t \rangle \\ &\leq 2\eta_t \|\mathbf{w}_t - \mathbf{w}^*\|_2^2 - 2\eta_{t+1} \|\mathbf{w}_{t+1} - \mathbf{w}^*\|_2^2 + 6\eta_t^3 \|\mathbf{b}_t\|_2^2 + 4\eta_t^2 \frac{r_{it}}{a_{it}\rho + (1-a_{it})/2} \phi(a_{it} \mathbf{w}^{*\top} x_{it}) \\ &\quad + 3c_{\alpha, \tilde{M}, 2} \eta_t^{\frac{3-\alpha}{1-\alpha}} + 4\eta_t^2 \langle \mathbf{w}^* - \mathbf{w}_t, \mathbf{b}_t \rangle, \end{aligned}$$

where we assume $\eta_t \geq \eta_{t+1}$ for every $t \in [T-1]$. Summing over j and taking into account that $\mathbf{w}_1 = \mathbf{0}$, we conclude

$$\begin{aligned} \sum_{j=1}^t \eta_j^2 \frac{r_{ij}}{a_{ij}\rho + (1-a_{ij})/2} \phi(a_{ij} \mathbf{w}_j^\top x_{ij}) &\leq 2\eta_1 \|\mathbf{w}^*\|_2^2 + \sum_{j=1}^t (6\eta_j^3 \|\mathbf{b}_j\|_2^2 \\ &+ 4\eta_j^2 \frac{r_{ij}}{a_{ij}\rho + (1-a_{ij})/2} \phi(a_{ij} \mathbf{w}^{*\top} x_{ij}) + 3c_{\alpha, \widetilde{M}, 2} \eta_j^{\frac{3-\alpha}{1-\alpha}} + 4\eta_j^2 \langle \mathbf{w}^* - \mathbf{w}_j, \mathbf{b}_j \rangle). \end{aligned} \quad (25)$$

As $x \mapsto x^{\frac{2a}{1+a}}$ is a concave function, we can apply Jensen's inequality and obtain

$$\begin{aligned} &\sum_{j=1}^t \eta_j^2 \left(\frac{r_{ij}}{a_{ij}\rho + (1-a_{ij})/2} \phi(a_{ij} \mathbf{w}_j^\top x_{ij}) \right)^{\frac{2\alpha}{1+\alpha}} \\ &\leq \sum_{j=1}^t \eta_j^2 \left(\frac{\sum_{j=1}^t \eta_j^2 \frac{r_{ij}}{a_{ij}\rho + (1-a_{ij})/2} \phi(a_{ij} \mathbf{w}_j^\top x_{ij})}{\sum_{j=1}^t \eta_j^2} \right)^{\frac{2\alpha}{1+\alpha}} \\ &= \left(\sum_{j=1}^t \eta_j^2 \right)^{\frac{1-\alpha}{1+\alpha}} \left[\sum_{j=1}^t \eta_j^2 \frac{r_{ij}}{a_{ij}\rho + (1-a_{ij})/2} \phi(a_{ij} \mathbf{w}_j^\top x_{ij}) \right]^{\frac{2\alpha}{1+\alpha}} \\ &\leq \left(\sum_{j=1}^t \eta_j^2 \right)^{\frac{1-\alpha}{1+\alpha}} \left[2\eta_1 \|\mathbf{w}^*\|_2^2 + \sum_{j=1}^t (6\eta_j^3 \|\mathbf{b}_j\|_2^2 + 4\eta_j^2 \frac{r_{ij}}{a_{ij}\rho + (1-a_{ij})/2} \phi(a_{ij} \mathbf{w}^{*\top} x_{ij}) \right. \\ &\quad \left. + 3c_{\alpha, \widetilde{M}, 2} \eta_j^{\frac{3-\alpha}{1-\alpha}} + 4\eta_j^2 \langle \mathbf{w}^* - \mathbf{w}_j, \mathbf{b}_j \rangle) \right]^{\frac{2\alpha}{1+\alpha}}. \end{aligned}$$

By substituting this into (24), and observing that $\mathbb{E}[\mathcal{E}_S(\mathbf{w}^*)] = \mathcal{E}(\mathbf{w}^*)$, we arrive at the following result

$$\begin{aligned} &\sum_{j=1}^t \eta_j \mathbb{E}[\mathcal{E}_S(\mathbf{w}_j) - \mathcal{E}_S(\mathbf{w}^*)] \\ &\leq \frac{1}{2} \|\mathbf{w}^*\|_2^2 + \frac{3}{2} \sum_{j=1}^t \eta_j^2 \sigma^2 d + \frac{3}{4} c_{\alpha, 1}^2 \widetilde{M}^{\frac{2}{1+\alpha}} \left(\sum_{j=1}^t \eta_j^2 \right)^{\frac{1-\alpha}{1+\alpha}} \mathbb{E} \left[2\eta_1 \|\mathbf{w}^*\|_2^2 \right. \\ &\quad \left. + \sum_{j=1}^t (6\eta_j^3 \|\mathbf{b}_j\|_2^2 + 4\eta_j^2 \frac{r_{ij}}{a_{ij}\rho + (1-a_{ij})/2} \phi(a_{ij} \mathbf{w}^{*\top} x_{ij}) + 3c_{\alpha, \widetilde{M}, 2} \eta_j^{\frac{3-\alpha}{1-\alpha}} + 4\eta_j^2 \langle \mathbf{w}^* - \mathbf{w}_j, \mathbf{b}_j \rangle) \right]^{\frac{2\alpha}{1+\alpha}} \\ &\leq \frac{1}{2} \|\mathbf{w}^*\|_2^2 + \frac{3}{2} \sum_{j=1}^t \eta_j^2 \sigma^2 d + \frac{3}{4} c_{\alpha, 1}^2 \widetilde{M}^{\frac{2}{1+\alpha}} \left(\sum_{j=1}^t \eta_j^2 \right)^{\frac{1-\alpha}{1+\alpha}} \left[2\eta_1 \|\mathbf{w}^*\|_2^2 + \sum_{j=1}^t (6\eta_j^3 \sigma^2 d \right. \\ &\quad \left. + 4\eta_j^2 \mathcal{E}(\mathbf{w}^*) + 3c_{\alpha, \widetilde{M}, 2} \eta_j^{\frac{3-\alpha}{1-\alpha}}) \right]^{\frac{2\alpha}{1+\alpha}}. \end{aligned}$$

Here we apply Jensen's inequality for concave functions, along with the fact that $\mathbb{E}_{\mathcal{M}}[\langle \mathbf{w}^* - \mathbf{w}_j, \mathbf{b}_j \rangle] = 0$ in the last step. This completes the proof. $\square$

6.2.3 Proof of Theorem 2 and Theorem 3

Proof of Theorem 2 The proof of the first part of Theorem 2, concerning the convergence of the excess weighted generalization error, follows a similar approach to that of Theorem 4 in Wang et al. (2024), and we will provide a summary here for the sake of completeness.

Recall that $\mathbf{w}_{\text{priv}} = \frac{1}{T} \sum_{t=1}^T \mathbf{w}_t$. As we are considering the case when $\eta_t = \eta$, we apply Jensen's inequality to obtain

$$\begin{aligned} \mathbb{E}[\mathcal{E}(\mathbf{w}_{\text{priv}})] - \mathcal{E}(\mathbf{w}^*) &\leq \left(\sum_{t=1}^T \eta_t \right)^{-1} \sum_{t=1}^T \eta_t \mathbb{E}[\mathcal{E}(\mathbf{w}_t) - \mathcal{E}(\mathbf{w}^*)] \\ &= \left(\sum_{t=1}^T \eta_t \right)^{-1} \sum_{t=1}^T \eta_t \mathbb{E}[\mathcal{E}(\mathbf{w}_t) - \mathcal{E}_S(\mathbf{w}_t)] + \left(\sum_{t=1}^T \eta_t \right)^{-1} \sum_{t=1}^T \eta_t \mathbb{E}[\mathcal{E}_S(\mathbf{w}_t) - \mathcal{E}(\mathbf{w}^*)]. \end{aligned} \quad (26)$$

From Proposition 1, we can bound the first term on the right-hand side of (26) as follows,

$$\begin{aligned} &\left(\sum_{t=1}^T \eta_t \right)^{-1} \sum_{t=1}^T \eta_t \mathbb{E}[\mathcal{E}(\mathbf{w}_t) - \mathcal{E}_S(\mathbf{w}_t)] \\ &= \mathcal{O}\left(\left(\kappa^{-1} T^{\frac{-2\alpha}{1+\alpha}} \eta^{\frac{-4\alpha}{1+\alpha}} + \kappa n^{-2} T^{\frac{2}{1+\alpha}} \eta^{\frac{2-2\alpha}{1+\alpha}}\right)(\eta + T\eta^3 \sigma^2 d + T\eta^2 \mathcal{E}(\mathbf{w}^*)\right. \\ &\quad \left.+ T\eta^{\frac{3-\alpha}{1-\alpha}} \eta^{\frac{2\alpha}{1+\alpha}} + \kappa^{\frac{1+\alpha}{\alpha-1}} + \kappa T\eta^{\frac{2}{1-\alpha}}\right). \end{aligned} \quad (27)$$

For the second term on the right-hand side of (26), Proposition 2 tells us that

$$\begin{aligned} &\left(\sum_{t=1}^T \eta_t \right)^{-1} \sum_{t=1}^T \eta_t \mathbb{E}[\mathcal{E}_S(\mathbf{w}_t) - \mathcal{E}(\mathbf{w}^*)] = (T\eta)^{-1} \sum_{t=1}^T \eta_t \mathbb{E}[\mathcal{E}_S(\mathbf{w}_t) - \mathcal{E}_S(\mathbf{w}^*)] \\ &= \mathcal{O}\left(T^{\frac{-2\alpha}{1+\alpha}} \eta^{\frac{1-3\alpha}{1+\alpha}} \left(\eta + T\eta^3 \sigma^2 d + T\eta^2 \mathcal{E}(\mathbf{w}^*) + T\eta^{\frac{3-\alpha}{1-\alpha}}\right)^{\frac{2\alpha}{1+\alpha}} + \eta \sigma^2 d + \frac{1}{T\eta}\right). \end{aligned} \quad (28)$$

Substituting (27) and (28) back into (26) gives

$$\begin{aligned} &\mathbb{E}[\mathcal{E}(\mathbf{w}_{\text{priv}})] - \mathcal{E}(\mathbf{w}^*) \\ &= \mathcal{O}\left(\kappa^{\frac{1+\alpha}{\alpha-1}} + \kappa T\eta^{\frac{2}{1-\alpha}} + \left(\kappa n^{-2} T^{\frac{2}{1+\alpha}} \eta^{\frac{2-2\alpha}{1+\alpha}} + T^{\frac{-2\alpha}{1+\alpha}} \eta^{\frac{1-3\alpha}{1+\alpha}} + \kappa^{-1} T^{\frac{-2\alpha}{1+\alpha}} \eta^{\frac{-4\alpha}{1+\alpha}}\right)\right. \\ &\quad \left.\times \left(\eta + T\eta^3 \sigma^2 d + T\eta^2 \mathcal{E}(\mathbf{w}^*) + T\eta^{\frac{3-\alpha}{1-\alpha}}\right)^{\frac{2\alpha}{1+\alpha}} + \eta \sigma^2 d + \frac{1}{T\eta}\right). \end{aligned} \quad (29)$$

Assuming $\eta T \geq 1$, we set $\eta = \min \left\{ \frac{c}{\sqrt{n}}, \frac{c\varepsilon}{\sqrt{d \log(1/\delta)}} \right\}$ and $\kappa = \sqrt{n}$. Given that

$$\sigma^2 d = \mathcal{O} \left(\frac{T d \log(1/\delta)}{n^2 \varepsilon^2} \right), \text{ we find}$$

$$\eta + T\eta^3 \sigma^2 d + T\eta^2 + T\eta^{\frac{3-\alpha}{1-\alpha}} = \mathcal{O} \left(\frac{T^2 \eta^3 d \log(1/\delta)}{n^2 \varepsilon^2} + T\eta^2 \right) = \mathcal{O} (T^2 n^{-2} \eta + T\eta^2).$$

Therefore, we derive

$$\begin{aligned} & \mathbb{E} [\mathcal{E}(\mathbf{w}_{\text{priv}})] - \mathcal{E}(\mathbf{w}^*) \\ &= \mathcal{O} \left(\left(T^{\frac{-2\alpha}{1+\alpha}} \eta^{\frac{1-3\alpha}{1+\alpha}} + n^{-\frac{1}{2}} \eta^{\frac{-4\alpha}{1+\alpha}} T^{\frac{-2\alpha}{1+\alpha}} + n^{-\frac{3}{2}} \eta^{\frac{2-2\alpha}{1+\alpha}} T^{\frac{2}{1+\alpha}} \right) \right. \\ & \quad \times (T^2 n^{-2} \eta + T\eta^2)^{\frac{2\alpha}{1+\alpha}} + n^{\frac{1+\alpha}{2(\alpha-1)}} + \frac{\sqrt{d \log(1/\delta)}}{n\varepsilon} + \frac{1}{\sqrt{n}} \Big). \end{aligned}$$

Assuming $T \asymp n$, we can conclude that for any $\alpha \in [1/2, 1)$,

$$\mathbb{E} [\mathcal{E}(\mathbf{w}_{\text{priv}})] - \mathcal{E}(\mathbf{w}^*) = \mathcal{O} \left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \log(1/\delta)}}{n\varepsilon} \right).$$

For $\alpha \in [0, 1/2)$, we define $\eta = \min \left\{ c n^{\frac{3(\alpha-1)}{2(1+\alpha)}}, \frac{c\varepsilon}{\sqrt{d \log(1/\delta)}} \right\} \leq \min \left\{ \frac{2}{\widetilde{ML}}, 1 \right\}$ and $\kappa = \sqrt{n}$

. This choice of η yields

$$\eta + T\eta^3 \sigma^2 d + T\eta^2 + T\eta^{\frac{3-\alpha}{1-\alpha}} = \mathcal{O} \left(\frac{T^2 \eta^3 d \log(1/\delta)}{n^2 \varepsilon^2} + T\eta^2 \right) = \mathcal{O} \left(\frac{T^2 \eta^2 \sqrt{d \log(1/\delta)}}{n(n\varepsilon)} + T\eta^2 \right).$$

With $T \asymp n^{\frac{2-\alpha}{1+\alpha}}$, combining this with (29) gives $\mathbb{E} [\mathcal{E}(\mathbf{w}_{\text{priv}})] - \mathcal{E}(\mathbf{w}^*) = \mathcal{O} \left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \log(1/\delta)}}{n\varepsilon} \right)$. This concludes the proof of the first part. The second part then follows directly by applying the comparison theorem in (5). $\square$

Proof of Theorem 3 By part 2 of Lemma 2, applied to the function $g(\mathbf{w}) := \phi(\mathbf{a}\mathbf{w}^\top x)$ and its Moreau envelope $g_\zeta(\mathbf{w}) := \phi^\zeta(\mathbf{a}\mathbf{w}^\top x)$, which states

$$\phi^\zeta(\mathbf{a}\mathbf{w}^\top x) \leq \phi(\mathbf{a}\mathbf{w}^\top x) \leq \phi^\zeta(\mathbf{a}\mathbf{w}^\top x) + \frac{G^2}{2\zeta}, \quad (30)$$

where G is the Lipschitz constant of ϕ . Recall that $0 < R \leq M$. Multiplying both sides of (30) by the weight $\frac{R}{A\rho + (1-A)/2}$, and taking expectation over (X, A, R) , we obtain

$$\begin{aligned} \mathcal{E}^{\phi^\zeta}(\mathbf{w}) &\leq \mathcal{E}(\mathbf{w}) \\ &\leq \mathcal{E}^{\phi^\zeta}(\mathbf{w}) + \frac{G^2}{2\zeta} \mathbb{E} \left[\frac{R}{A\rho + (1-A)/2} \right] \\ &= \mathcal{E}^{\phi^\zeta}(\mathbf{w}) + \frac{G^2}{2\zeta} \int_{\mathcal{X}} \{ \mathbb{E}[R|X=x, A=1] + \mathbb{E}[R|X=x, A=-1] \} dP_{\mathcal{X}} \\ &\leq \mathcal{E}^{\phi^\zeta}(\mathbf{w}) + \frac{MG^2}{\zeta} \end{aligned}$$

for all $\mathbf{w}$. Let $\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathbb{R}^d} \mathcal{E}^{\phi^\zeta}(\mathbf{w})$. Using the fact that $\mathcal{E}^{\phi^\zeta}(\hat{\mathbf{w}}) \leq \mathcal{E}^{\phi^\zeta}(\mathbf{w}^*)$ by the definition of $\hat{\mathbf{w}}$, we obtain

$$\begin{aligned} \mathbb{E}[\mathcal{E}(\mathbf{w}_{\text{priv}})] - \mathcal{E}(\mathbf{w}^*) &\leq \mathbb{E}[\mathcal{E}^{\phi^\zeta}(\mathbf{w}_{\text{priv}})] - \mathcal{E}^{\phi^\zeta}(\mathbf{w}^*) + \frac{MG^2}{\zeta} \\ &\leq \mathbb{E}[\mathcal{E}^{\phi^\zeta}(\mathbf{w}_{\text{priv}})] - \mathcal{E}^{\phi^\zeta}(\hat{\mathbf{w}}) + \frac{MG^2}{\zeta}. \end{aligned}$$

Furthermore, since ϕ^ζ is ζ -smooth and $2G$ -Lipschitz by part 1 of Lemma 2, we can apply Theorem 2 to obtain

$$\mathbb{E}[\mathcal{E}^{\phi^\zeta}(\mathbf{w}_{\text{priv}})] - \mathcal{E}^{\phi^\zeta}(\hat{\mathbf{w}}) = \mathcal{O} \left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \log(1/\delta)}}{n\varepsilon} \right)$$

by setting $T \asymp n$ and $\eta_t = c \min \left\{ \frac{1}{\sqrt{n}}, \frac{\varepsilon}{\sqrt{d \log(1/\delta)}} \right\} \leq \min \left\{ \frac{2}{\tilde{M}_\zeta}, 1 \right\}$ for some constant $c > 0$. Combining the above two inequalities, we have

$$\mathbb{E}[\mathcal{E}(\mathbf{w}_{\text{priv}})] - \mathcal{E}(\mathbf{w}^*) \leq \mathcal{O} \left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \log(1/\delta)}}{n\varepsilon} \right) + \frac{MG^2}{\zeta}.$$

Finally, since ζ is chosen to satisfy

$$\frac{MG^2}{\zeta} = \mathcal{O} \left(\frac{1}{\sqrt{n}} + \frac{\sqrt{d \log(1/\delta)}}{n\varepsilon} \right),$$

we complete the proof. $\square$

Table 2 Average misclassification rate and empirical treatment value under different ε and sampling rates γ

ε	γ	Average misclassification rate	Empirical treatment value
0.1	0.001	0.4952 (0.1299)	7.8177 (0.9093)
	0.002	0.4877 (0.1336)	7.8474 (0.9586)
	0.005	0.4946 (0.1223)	7.7931 (0.8978)
	0.010	0.4819 (0.1447)	7.8221 (1.0692)
	0.020	0.4877 (0.1368)	7.7828 (0.9987)
	0.050	0.4830 (0.1367)	7.8541 (0.9902)
	0.100	0.4951 (0.1434)	7.8015 (0.9688)
	0.200	0.4882 (0.1373)	7.8427 (0.9487)
	0.500	0.4835 (0.1434)	7.8985 (1.0262)
	1.000	0.4851 (0.1283)	7.8919 (0.9150)
3	0.001	0.2044 (0.1056)	9.4455 (0.6891)
	0.002	0.1973 (0.1006)	9.4685 (0.6857)
	0.005	0.1917 (0.1005)	9.5078 (0.6462)
	0.010	0.1923 (0.0876)	9.5174 (0.6230)
	0.020	0.1904 (0.0985)	9.5106 (0.6740)
	0.050	0.2028 (0.1118)	9.4515 (0.7365)
	0.100	0.1881 (0.0955)	9.5232 (0.6260)
	0.200	0.1952 (0.1075)	9.5654 (0.6935)
	0.500	0.1948 (0.1019)	9.5782 (0.6687)
	1.000	0.1828 (0.0912)	9.5413 (0.6275)
6	0.001	0.1249 (0.0533)	9.7154 (0.4530)
	0.002	0.1193 (0.0530)	9.7289 (0.4449)
	0.005	0.1231 (0.0540)	9.7198 (0.4548)
	0.010	0.1154 (0.0522)	9.7389 (0.4526)
	0.020	0.1163 (0.0478)	9.7389 (0.4329)
	0.050	0.1184 (0.0485)	9.7315 (0.4490)
	0.100	0.1178 (0.0528)	9.7321 (0.4592)
	0.200	0.1166 (0.0535)	9.7339 (0.4587)
	0.500	0.1208 (0.0538)	9.7306 (0.4733)
	1.000	0.1196 (0.0496)	9.7291 (0.4681)
∞	0.001	0.0776 (0.0345)	9.9412 (0.4226)
	0.002	0.0725 (0.0279)	9.9530 (0.4266)
	0.005	0.0679 (0.0263)	9.9579 (0.4236)
	0.010	0.0664 (0.0264)	9.9598 (0.4220)
	0.020	0.0668 (0.0255)	9.9603 (0.4242)
	0.050	0.0650 (0.0241)	9.9636 (0.4241)
	0.100	0.0647 (0.0243)	9.9645 (0.4249)
	0.200	0.0648 (0.0244)	9.9623 (0.4247)
	0.500	0.0649 (0.0246)	9.9629 (0.4248)
	1.000	0.0648 (0.0244)	9.9631 (0.4246)

All values are reported as mean (SD) across repeated runs. Here $\varepsilon = \infty$ represents the non-private mini-batch SGD baseline

Supplementary Experiments

We extend the privacy analysis to uniform without-replacement mini-batch sampling with sampling rate $\gamma = k/n$ ($1 \leq k < n$), which satisfies the assumption of Lemma 3. The gradient update becomes

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta_t \left(\frac{1}{k} \sum_{i \in \mathcal{B}_t} \frac{r_i}{a_i \rho + (1 - a_i)/2} \partial \phi(a_i \mathbf{w}_t^\top \mathbf{x}_i) + \mathbf{b}_t \right), \quad (31)$$

where $\mathcal{B}_t$ is a mini-batch of size k sampled without replacement, and $\mathbf{b}_t \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ with

$$\sigma^2 = \frac{14\tilde{M}^2 G^2 T}{\beta n^2 \varepsilon} \left(\frac{\log(1/\delta)}{(1 - \beta)\varepsilon} + 1 \right).$$

For the without-replacement subsampling scheme, the ℓ_2 -sensitivity of the batch-averaged gradient is upper bounded by $2\tilde{M}G/k$. Following the proof of Theorem 1, we derive the formal privacy guarantee for Algorithm (31) as stated below.

Theorem 5 (*Privacy Guarantee of mini-batch DP-SGD*) *When the convex loss function is G -Lipschitz continuous with $G > 0$, Algorithm (31) guarantees (ε, δ) -DP for some $\beta \in (0, 1)$*

if $\sigma^2 \geq 2.68\tilde{M}^2 G^2 / k^2$ and $\lambda - 1 \leq \frac{\sigma^2 k^2}{6\tilde{M}^2 G^2} \log \left(\frac{n}{\lambda k \left(1 + \frac{\sigma^2 k^2}{4\tilde{M}^2 G^2} \right)} \right)$, where $\lambda = \frac{\log(1/\delta)}{(1 - \beta)\varepsilon} + 1$.

Remark 4 From the closed-form expression of σ^2 , the adjusted noise variance is independent of both the batch size k and the sampling rate $\gamma = k/n$. This occurs because two opposing effects exactly cancel out in the calculation: subsampling broadens the privacy protection range, while averaging gradients across batches reduces gradient sensitivity. Specifically, $\gamma^2 \cdot \Delta_g^2 = \left(\frac{k}{n}\right)^2 \cdot \left(\frac{2\tilde{M}G}{k}\right)^2 = \frac{4\tilde{M}^2 G^2}{n^2}$, so k no longer appears in the privacy budget calculation. Increasing k greatly relaxes the two feasibility constraints from Theorem 5, yet the magnitude of the DP noise remains unchanged when ε, δ, T are fixed. From an optimization perspective, changing the sampling rate only affects the natural random error of the mini-batch gradient, not the magnitude of the DP noise. A higher sampling rate reduces gradient fluctuations and thereby diminishes the relative impact of a fixed DP noise level during parameter updates.

We conduct numerical simulations with the following setup: sample size $n = 1000$, step size $\eta = 0.01$, iteration count $T = 1500$, and 200 repeated runs. We use the logistic loss $\phi_{\log}$ and generate data in the same way as Sect. 5.1. We evaluate the average misclassification rate and the empirical treatment value under different privacy parameters $\varepsilon \in \{0.1, 3, 6, \infty\}$ and sampling rates $\gamma \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1\}$. We report the mean and standard deviation (SD) of these two metrics. Table 2 shows that, for a fixed privacy parameter ε , larger sampling rates γ lead to slight performance improvement. Both misclassification rate and empirical treatment value change very little across different γ . This is consistent with our theoretical analysis: although the DP noise magnitude is inde-

pendent of γ , a larger sampling rate reduces gradient stochastic fluctuation. It yields small performance gains and more stable outputs. In addition, a larger ε substantially reduces the misclassification rate and improves the empirical treatment value. The non-private case $\varepsilon = \infty$ has the best and most stable performance.

Acknowledgements The work by J. Fan is partially supported by the Research Grants Council of Hong Kong [Project No. HKBU 12302923 and 12303024], Guangdong Basic and Applied Basic Research Fund [Project No. 2024A1515011878], Guangdong and Hong Kong Universities “1+1+1” Joint Research Collaboration Scheme [Project No. 2025A0505000007], and Hong Kong Baptist University [Project No. RC-FNRA-IG/22-23/SCI/02]. The work by Y. Lei is partially supported by the Research Grants Council of Hong Kong [Project No. 17302624]. The work by D. H. Xiang is supported by the National Natural Science Foundation of China (Grant No. 12471487 and 62536006).

Author Contributions This work was carried out through the joint efforts of all authors. A.Y. is a graduate student and contributed to the development of the manuscript under the collective supervision of J.F., Y.L., and D.X., who all served as principal advisors throughout the research. All authors participated in problem formulation, methodological design, experimental verification, and manuscript preparation. All authors reviewed and approved the final version of the manuscript.

Funding Open access funding provided by Hong Kong Baptist University Library

Data Availability This study involves both synthetic and publicly available data. The simulation experiments were conducted using synthetic data generated by the authors for demonstration purposes. The real-world clinical dataset used in the evaluation is publicly available from the UCI Machine Learning Repository: AIDS Clinical Trials Group Study 175.

Declarations

Conflict of interest The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article’s Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article’s Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Bassily, R., Feldman, V., Guzmán, C., & Talwar, K. (2020). Stability of stochastic gradient descent on nonsmooth convex losses. *Advances in Neural Information Processing Systems*, 33, 4381–4391.
- Bassily, R., Feldman, V., Talwar, K., & Thakurta, A. G. (2019). Private stochastic convex optimization with optimal rates. *Advances in Neural Information Processing Systems*, 32, 11279–11288.
- Bassily, R., Guzmán, C., & Menart, M. (2021). Differentially private stochastic optimization: New results in convex and non-convex settings. *Advances in Neural Information Processing Systems*, 34, 9317–9329.
- Bassily, R., Smith, A., & Thakurta, A. (2014). Private empirical risk minimization: Efficient algorithms and tight error bounds. In *Proceedings of the 55th IEEE Annual Symposium on Foundations of Computer Science (FOCS)* (pp. 464–473). Philadelphia, PA, USA: IEEE.
- Bousquet, O., & Elisseeff, A. (2002). Stability and generalization. *Journal of Machine Learning Research*, 2, 499–526.
- Chen, X., Tang, B., Fan, J., & Guo, X. (2022). Online gradient descent algorithms for functional data learning. *Journal of Complexity*, 70, Article 101635.

- Choi, T., Lee, H., & Choi, S. (2023). Accountable survival contrast-learning for optimal dynamic treatment regimes. *Scientific Reports*, 13(1), 2250.
- Cucker, F., & Zhou, D. X. (2007). Learning Theory: An Approximation Theory Viewpoint. *Cambridge Monographs on Applied and Computational Mathematics*. Cambridge University Press, Cambridge, UK
- Dwork, C., & Lei, J. (2009). Differential privacy and robust statistics. In *Proceedings of the 41st Annual ACM Symposium on Theory of Computing* (pp. 371–380).
- Dwork, C., McSherry, F., Nissim, K., & Smith, A. (2006). Calibrating noise to sensitivity in private data analysis. In *Proceedings of the Third Theory of Cryptography Conference (TCC 2006)*, (pp. 265–284).
- Dwork, C., & Roth, A. (2014). The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9, 211–407.
- Fan, J., Guo, Z. C., & Shi, L. (2024). Spectral algorithms for functional linear regression. *Communications on Pure and Applied Analysis*, 23(7), 895–915.
- Fan, J., & Lei, Y. (2024). High-probability generalization bounds for pointwise uniformly stable algorithms. *Applied and Computational Harmonic Analysis*, 70, Article 101632.
- Fan, J., Lv, F., & Shi, L. (2019). An RKHS approach to estimate individualized treatment rules based on functional predictors. *Mathematical Foundations of Computing*, 2(2), 169–181.
- Fan, J., & Xiang, D. H. (2020). Quantitative convergence analysis of kernel based large-margin unified machines. *Communications on Pure and Applied Analysis*, 19(8), 4069–4083.
- Feldman, V., Koren, T., & Talwar, K. (2020). Private stochastic convex optimization: Optimal rates in linear time. In *Proceedings of the 52nd Annual ACM SIGACT Symposium on Theory of Computing* (pp. 439–449).
- Fu, S., He, Q. Y., Zhang, S. G., & Liu, Y. F. (2019). Robust outcome weighted learning for optimal individualized treatment rules. *Journal of Biopharmaceutical Statistics*, 29(4), 606–624.
- Giddens, S., Zhou, Y., Krull, K. R., Brinkman, T. M., Song, P. X., & Liu, F. (2023). A differentially private weighted empirical risk minimization procedure and its application to outcome weighted learning. [ArXiv:2307.13127](https://arxiv.org/abs/2307.13127)
- Hammer, S. M., Katzenstein, D. A., & Hughes, M. D. (1996). A trial comparing nucleoside monotherapy with combination therapy in hiv-infected adults with cd4 cell counts from 200 to 500 per cubic millimeter. aids clinical trials group study 175 team. *The New England Journal of Medicine*, 335(15), 1081–1090.
- Hardt, M., Recht, B., & Singer, Y. (2016). Train faster, generalize better: Stability of stochastic gradient descent. In *International Conference on Machine Learning (ICML)* (pp. 1225–1234).
- Jiang, D., Sun, S., & Yu, Y. (2023). Functional rényi differential privacy for generative modeling. In *Proceedings of the 37th International Conference on Neural Information Processing Systems (NeurIPS)*, (pp. 14797–14817).
- Jiang, R., Lu, W., Song, R., & Davidian, M. (2017). On estimation of optimal treatment regimes for maximizing t-year survival probability. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 79(4), 1165–1185.
- Kuzborskij, I., & Lampert, C. (2018). Data-dependent stability of stochastic gradient descent. In *International Conference on Machine Learning (ICML)* (pp. 2820–2829).
- Lee, J., & Chen, G. (2026). Differentially private two-stage empirical risk minimization and applications to individualized treatment rule [arXiv:2602.12604](https://arxiv.org/abs/2602.12604)
- Lei, Y., & Ying, Y. (2020). Fine-grained analysis of stability and generalization for stochastic gradient descent. In *International Conference on Machine Learning (ICML)* (pp. 5809–5819).
- Liang, Z., Wang, B., Gu, Q., Osher, S., & Yao, Y. (2020). Exploring private federated learning with Laplacian smoothing [arXiv:2005.00218](https://arxiv.org/abs/2005.00218)
- Mancinelli, L., Cronin, M., & Sadee, W. (2000). Pharmacogenomics: The promise of personalized medicine. *AAPS (Pharmaceutical Sciences)*, 2, 29–41.
- Mironov, I. (2017). Rényi differential privacy. In *Proceedings of the 30th IEEE Computer Security Foundations Symposium (CSF)*. IEEE (pp. 263–275). Santa Barbara, CA, USA.
- Nesterov, Y. (2005). Smooth minimization of non-smooth functions. *Mathematical Programming*, 103(1), 127–152.
- Qi, Z. L., & Liu, Y. F. (2018). D-learning to estimate optimal individual treatment rules. *Electronic Journal of Statistics*, 12(2), 3601–3638.
- Qian, M., & Murphy, S. A. (2011). Performance guarantees for individualized treatment rules. *The Annals of Statistics*, 39, 1180–1210.
- Spicker, D., Moodie, E. E. M., & Shortreed, S. M. (2024). Differentially private outcome-weighted learning for optimal dynamic treatment regime estimation. *Stat*, 13(1), Article e641.
- Su, J., Hu, L., & Wang, D. (2022). Faster rates of private stochastic convex optimization. In *Proceedings of the International Conference on Algorithmic Learning Theory (ALT)* (pp. 995–1002).
- Wang, C., & Fan, J. (2024). On the convergence of gradient descent for robust functional linear regression. *Journal of Complexity*, 84, Article 101858.

- Wang, J., Liu, Q., Liang, H., Joshi, G., & Poor, H. V. (2021a). A novel framework for the analysis and design of heterogeneous federated learning. *IEEE Transactions on Signal Processing*, 69, 5234–5249.
- Wang, P., Lei, Y., Ying, Y., & Zhang, H. (2022). Differentially private SGD with non-smooth losses. *Applied and Computational Harmonic Analysis*, 56, 306–336.
- Wang, P., Lei, Y., Ying, Y., & Zhou, D. X. (2024). Differentially private stochastic gradient descent with low-noise. *Neurocomputing*, 585, Article 127557.
- Wang, P., Yang, Z., Lei, Y., Ying, Y., & Zhang, H. (2021b). Differentially private empirical risk minimization for auc maximization. *Neurocomputing*, 461, 419–437.
- Xue, Z., Yang, S., Huai, M., & Wang, D. (2021). Differentially private pairwise learning revisited. In *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)* (pp. 3242–3248).
- Yang, A. L., Fan, J., & Xiang, D. H. (2025). Online outcome weighted learning with general loss functions. *Journal of Complexity*, 88, Article 101931.
- Yang, Z., Lei, Y., Wang, P., Yang, T., & Ying, Y. (2021). Simple stochastic and online gradient descent algorithms for pairwise learning. In *Proceedings of the 35th International Conference on Neural Information Processing Systems (NeurIPS)*, (pp. 1–12).
- Zhang, B., Tsiatis, A. A., Laber, E. B., & Davidian, M. (2012). A robust method for estimating optimal treatment regimes. *Biometrics*, 68, 1010–1018.
- Zhang, X., Hong, M., Dhople, S. V., Yin, W., & Liu, Y. (2021). Fedpd: A federated learning framework with adaptivity to non-iid data. *IEEE Transactions on Signal Processing*, 69, 6055–6070.
- Zhao, Y. Q., Zeng, D. L., Rush, A. J., & Kosorok, M. R. (2012). Estimating individualized treatment rules using outcome weighted learning. *Journal of the American Statistical Association*, 107(499), 1106–1118.
- Zhu, X. X., & Xiang, D. H. (2022). Error analysis of OWL algorithms with varying gaussians and convex loss. Zhejiang Normal University. Master's thesis.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Aoli Yang¹ · Jun Fan² · Yunwen Lei³ · Dao-Hong Xiang¹

✉ Jun Fan
junfan@hkbu.edu.hk

Aoli Yang
al.yang@zjnu.edu.cn

Yunwen Lei
leiyw@hku.hk

Dao-Hong Xiang
daohongxiang@zjnu.cn

¹ School of Mathematical Science, Zhejiang Normal University, Zhejiang Key Laboratory of Intelligent Education Technology and Application, Jinhua 321004, Zhejiang, China

² Department of Mathematics, Hong Kong Baptist University, Kowloon, Hong Kong, China

³ Department of Mathematics, The University of Hong Kong, Pokfulam, Hong Kong, China